

READING 30

INTRODUCTION TO COMMODITIES AND COMMODITY DERIVATIVES

EXAM FOCUS

This topic review will help you understand different commodity sectors and key factors influencing prices in those sectors. Pay special attention to what backwardation and contango mean in terms of spot and futures prices. You should understand the different components of returns to commodity futures and what determines whether roll return is positive or negative. Finally, familiarize yourself with the Insurance Theory, the Hedging Pressure Hypothesis, and the Theory of Storage and what they say about futures prices.

MODULE 30.1: INTRODUCTION AND THEORIES OF RETURN



Video covering this content is available online.

LOS 30.a: Compare characteristics of commodity sectors.

Fundamental Analysis

Supply and demand shocks affect commodity prices. Fundamental analysis seeks to predict these shocks using a variety of inputs/tools such as the following:

- *Announcements.* Government or private agencies produce forecasts, which can be useful for analysts in their models.
- *Component analysis.* Different products that make up the commodity class are the components of that commodity (e.g., jet fuel, diesel, gasoline, and lubricants are components of crude oil). Analysis of demand and supply forecasts of components can fine-tune the forecasts for the aggregate.
- *Timing issues.* Incorporating any seasonality and previously observed logistical issues can refine estimates of demand/supply.
- *Macro.* Macroeconomic factors such as inflation may trigger an increase in demand for commodities. Low interest rates may increase capital spending and, therefore, the demand for commodities in the intermediate term.

Commodities can be classified by their characteristics into sectors, including:

- Energy—crude oil, natural gas, and refined petroleum products.
- Industrial metals—aluminum, nickel, zinc, lead, tin, iron, and copper.

- Grains—wheat, corn, soybeans, and rice.
- Livestock—hogs, sheep, cattle, and poultry.
- Precious metals—gold, silver, and platinum.
- Softs (cash crops)—coffee, sugar, cocoa, and cotton.

The factors that influence supply and demand and the nature of production differ for these sectors. A summary of these differences can help explain the differences in price dynamics among the sectors.

The **energy sector** comprises crude oil, natural gas, and refined products. It is the sector with the greatest market value and is a very important source of revenue to many countries and regions.

Crude oil from different regions has different characteristics. Light oil (low viscosity) and sweet oil (low sulfur content) are less costly to refine and, therefore, sell at a premium relative to heavier or higher sulfur crude oils. Crude oil can be stored indefinitely by keeping it in the ground and is also stored in tanks and aboard tanker ships. Many countries store large amounts of crude oil as strategic reserves.

The supply of crude oil has been augmented by advances in drilling and extraction technology, especially in the 21st century. While global economic growth is an important driver of worldwide demand for oil, other factors have slowed this growth in demand. Improvements in refining technology have tended to increase the output of petroleum distillates from each barrel of crude oil, and improved engines are able to produce more work from each gallon of these distillates.

Economic cycles also affect the demand for oil, which is higher during expansions when credit is widely available and can decrease sharply when contractions lead to reductions in the availability of credit.

Improvements in the efficiency of alternative sources of energy production have also reduced the overall growth in the demand for oil. Increasingly stringent restrictions on oil exploration and production in response to environmental concerns have tended to increase the cost of oil production and decrease supply.

Political risk is an important factor in oil supply. Over half the crude oil supply comes from countries in the Middle East, and conflict there can reduce supply dramatically.

Refined products such as gasoline, heating oil, and jet fuel, are only stored for short periods. Refinery output is the relevant supply consideration. The geographic concentration of refinery capacity means that extreme weather in some coastal regions can significantly affect the supply of refined products.

Seasonal factors affect the demand for refined products in that greater vacation travel in the summer months increases gasoline demand, and colder weather in the winter increases the demand for heating oil.

Unlike crude oil, **natural gas** can be used just as it comes out of the ground with very little processing. Transportation costs play an important role in energy pricing. Crude oil can be transported at a relatively low cost on ships, while natural gas must be cooled to its liquid state to be transported by ship, significantly increasing the cost of transport.

The supply of *associated gas*—gas produced in conjunction with the extraction of crude oil—is tied to the production of crude oil. *Unassociated gas* is produced from formations where oil is not present so that its supply is not tied to the demand for and production of crude oil.

Worldwide demand and supply for gas depends on many of the same factors as supply and demand for crude oil, but seasonality due to weather is more pronounced. Cold winters increase the demand for gas for heating fuel. Hot summers increase the demand for gas as well (for cooling) because gas is a primary source of fuel for electrical power generation.

Demand for **industrial metals** is primarily tied to GDP growth and business cycles because these metals are used extensively in construction and manufacturing. Storage of metals is not costly.

Political factors, especially union strikes and restrictive environmental regulations, can have a significant effect on the supply of an industrial metal. Industrial metals must be smelted from mined ore. Both mines and smelters are large-scale operations with high development costs and high fixed costs.

Grains are grown over an annual cycle and stored, although multiple crops in a single year are possible in some areas. The risks to grain supply are the usual: droughts, hail, floods, pests, diseases, changes in climate, and so on. It would be difficult to overstate the importance of grains in feeding the world's population, especially given the potential for political instability when grain stocks are insufficient.

Precious metals are used in electronics and for jewelry and can be stored indefinitely. Gold has long been used as a store of value and has provided a hedge against the inflation risk of holding currency. Jewelry demand is high where wealth is being accumulated. Industrial demand for precious metals is sensitive to business cycles.

Livestock supply depends on the price of grain, which is the primary input in its production. When increasing grain prices increase the cost of feeding livestock, the rate of slaughter also increases, which leads to a decrease in price. Such a drawdown in population can result in subsequent increases in price over time.

Weather can affect the production of some animals. Disease is a source of significant risk to livestock producers, and some diseases have had a large impact on market prices.

Income growth in developing economies is an important source of growth in demand for livestock. Freezing allows the storage of meat products for a limited amount of time.

Softs refers to cotton, coffee, sugar, and cocoa, which are all grown in the warmer climates of the lower latitudes. Just as with grains, weather is the primary factor in determining production and price, but disease is a significant risk as well. Demand increases with increases in incomes in developing economies but is dependent on consumer tastes as well.

LOS 30.b: Compare the life cycle of commodity sectors from production through trading or consumption.

The life cycle of crude oil begins with the time it takes to drill a well and extract the crude. After being transported, crude oil is typically stored for no more than a few months. The next step is refining the crude oil into various fuels such as gasoline, heating oil, diesel oil, and jet fuel. These fuels must then be transported to the consumer.

Natural gas requires minimal processing after it is extracted. While natural gas often reaches the consumer through a pipeline, it can be cooled to liquid form and transported on specially constructed ships. Energy commodities are delivered year-round, but demand is seasonal to some extent.

The life cycle of industrial metals is straightforward: the extracted ore is smelted into the quality of metal that end users need. Industrial metals can be stored indefinitely in most cases, and the regular flow of output means that end users can meet their needs with monthly deliverable futures contracts.

A key characteristic of industrial metals production is economies of scale due to large, efficient mining and smelting operations. The large size projects required for efficient operation cost billions of dollars and take significant time to construct. Construction of new capacity or facilities when capacity utilization and earnings are high can result in the additional capacity coming online at or past the peak of the economic cycle. Mining and smelting operations are most efficient running near their capacity, so individual producers are hesitant to decrease production when prices fall because the peak of the economic cycle has passed or because facilities growth has created excess capacity in the industry.

Livestock production times vary with animal size, with chickens ready for slaughter after only weeks, hogs in about six months, and cattle after a few years. Freezing allows storage for some period after slaughter and allows international trade in livestock products such as frozen beef. Livestock production has a significant seasonal component.

Grain production is seasonal, so deliverable futures contracts are available on dates to coincide with the harvest. Because planting occurs five months or more before harvest, quantities harvested are set largely by expectations for demand when crops are planted. Grains can be stored for significant time periods after harvest. The six-month offset to harvest times in the northern and southern hemispheres brings crops to markets more frequently.

Production cycles and storage options for softs vary by product. Among softs, coffee offers an example of an agricultural commodity that is harvested somewhere around the world in almost every month. Coffee is stored in warehouses after transport by ship. Local coffee roasters then roast the beans and deliver to end users or to retail sales outlets. Coffee plants can take up to four years to produce the fruit that will become coffee beans, so there is a significant lag between investment in new capacity and increases in supply.

To hedge their price risk, coffee producers can sell in the futures market for delivery to a warehouse, and consumer companies can buy in the futures market and take delivery at the warehouse. Two different types of coffee beans are traded, Robusta and Arabica, with Arabica being the premium product.

LOS 30.c: Contrast the valuation of commodities with the valuation of equities and bonds.

Unlike stocks and bonds, commodities are physical assets, have no cash flows, and may incur storage and transportation costs.

Stocks and bonds (financial assets) can be valued by calculating the present value of their expected future cash flows (e.g., dividends, interest, etc.). Commodities produce no earnings or cash flows; however, the current (spot) price of a commodity can be viewed as the discounted value of the expected selling price at some future date. Storage costs for commodities can lead to forward prices that are higher the further the forward settlement date is in the future.

LOS 30.d: Describe types of participants in commodity futures markets.

Participants in commodity futures markets can be categorized as hedgers, traders and investors, exchanges, analysts, and regulators.

Traders and investors in the commodities market can be classified as informed investors—those who provide liquidity to the markets—and arbitrageurs. **Hedgers** are considered informed investors because they either produce or use the commodity. Hedgers reduce their risk by buying (going long) or selling (going short) futures contracts. A corn farmer can reduce the uncertainty about the price she will receive for her corn by selling corn futures. A cattle producer, however, would hedge his price risk by buying corn futures to reduce his uncertainty about the cost of feed for the cattle.



PROFESSOR'S NOTE

Hedgers are said to “do in the futures market what they must do in the future.” A wheat farmer will need to sell wheat in the future (i.e., after the harvest) and can hedge price risk by selling futures contracts. A grain miller will need to buy wheat in the future and can hedge price risk by buying futures contracts.

Speculators take on commodity risk in futures markets and may act as informed investors, seeking to exploit an information or information processing advantage to profit from trading with hedgers. Speculators can also earn profits by providing liquidity to markets: buying futures when short hedgers (commodity producers) are selling and selling futures when long hedgers (commodity users) are buying.

Arbitrageurs in the commodity markets are often those in the business of buying, selling, and storing the physical commodities when the difference between spot and futures prices is too large or too small based on the actual cost of storing the commodity. When the difference is too large, an arbitrageur can buy and store the commodity and sell it at its (too high) futures price. When the difference is too small, an arbitrageur can effectively “not store” the commodity by selling from his own inventory and going long futures, replacing the inventory at the future date.

Commodity exchanges operate in many of the world’s financial centers to reflect the worldwide production and consumption of commodities as well as the globalization of

financial markets in general. Investors can trade commodity futures on a smart phone or via a Bloomberg terminal.

Commodity market analysts, considered non-market participants, use market information to perform analytical work for various entities including governments, universities, economic forecasters, and commercial data analysis firms.

Various **commodity regulators** are responsible for the regulation of commodities markets around the world. In the U.S., the Commodities Futures Trading Commission (CFTC) is responsible for market regulation.

LOS 30.e: Analyze the relationship between spot prices and futures prices in markets in contango and markets in backwardation.

The difference between the spot (cash) market price and the futures price for a date in the future is referred to as the **basis** of that particular contract. The basis is calculated as the spot price minus the futures price and can be positive or negative. The difference between the futures price of a nearer maturity and the futures price of a more-distant maturity is known as the **calendar spread**.

When futures prices are higher at dates further in the future, the futures market (or futures curve) is said to be in **contango**. In a contango market, the calendar spread and basis are negative. Conversely, if futures prices are lower at dates further in the future, the market is said to be in **backwardation**, and the basis and calendar spread are positive.

When a futures market is in backwardation, long futures positions have a positive returns component (the “roll return,” which we will describe later in this topic review). With a futures curve in backwardation, futures prices are lower than spot prices for the commodity. Since futures prices converge to spot prices over the term of a futures contract, there is a positive returns component from the passage of time.

When a futures market is in contango, so that futures prices are greater than spot prices, there is a negative returns component for long futures positions. As time passes, convergence of futures prices to spot prices (or longer-dated futures prices to nearer-term futures prices) results in a decrease in the value of a long futures position.

LOS 30.f: Compare theories of commodity futures returns.

Three theories of the determinants of returns on commodities, based on the shape of the futures curve, have been expounded: the Insurance Theory, the Hedging Pressure Hypothesis, and the Theory of Storage.

Economist John Maynard Keynes put forward the **Insurance Theory** of futures returns, which states that the desire of commodity producers to reduce their price risk drives commodity futures returns. Producers face uncertainty about the price they will receive for their output and reduce this uncertainty by selling futures contracts. This selling drives down futures prices. The Insurance Theory states that the futures prices will be less than current spot prices to provide a return to those buying futures from producers (i.e., speculators). In this view, the resulting positive return to the buyers of futures contracts is

their return for providing insurance against price uncertainty to producers. Keynes contended that this results in backwardation “normally,” and the situation was termed “normal backwardation” based on this theory.

The Insurance Theory was found to be lacking based on two empirical findings. The first finding is that for markets in backwardation, buying futures has not resulted in the extra returns the theory says buyers should receive for providing “insurance.” The second finding is that many markets are not in backwardation but are in contango (future prices higher than spot prices), which would imply a negative return for providing insurance to producers.

The **Hedging Pressure Hypothesis** added the hedging behavior of commodity consumers to the Insurance Theory in an attempt to better explain observed futures returns. Just as a wheat farmer faces uncertainty about the price at which he will sell his wheat in the future, a baking company faces uncertainty about the price it will pay for flour in the future. To hedge its price risk, the baking company will go long wheat futures. The more commodity users hedge with long positions (buying futures), the more upward price pressure there is on the futures price. Under the Hedging Pressure Hypothesis, when producers’ hedging behavior dominates, the market will be in backwardation, and when users’ hedging behavior dominates, the market will be in contango.

Despite the intuitive appeal of the Hedging Pressure Hypothesis, it has some shortcomings. Producers typically face more concentrated price risk than consumers. Individual consumers will spend only a small portion of their income on a single commodity, and for commercial users of the commodity, the actual cost of the commodity may represent only a small portion of the total cost of the production.

Additionally, both producers and consumers may be speculators in the market, not just hedgers. Another problem with the Hedging Pressure Hypothesis is that hedging pressure is not observable, so we cannot directly test the hypothesis that relative hedging pressure is the cause of backwardation and contango.

The **Theory of Storage** is based on the idea that whether a futures market is in backwardation or contango depends on the relationship between the costs of storing the commodity for future use and the benefits of holding physical inventory of the commodity. When the costs of storage outweigh the benefits of holding physical inventory, futures are more attractive than current inventory, futures will trade at a higher price than spot, and the market will be in contango. Conversely, when the benefits of holding physical inventory outweigh the costs of storage, current possession is more attractive than future possession, spot prices are higher than futures prices, and the market will be in backwardation.

The benefits of having physical inventory available are referred to as a commodity’s **convenience yield**. When physical stocks are low and there is a high probability that the commodity will be in short supply, the benefits of holding physical stock (and the convenience yield) are higher.

The Theory of Storage takes both the costs and benefits of holding a commodity into account in the following relation:

$$\text{futures price} = \text{spot price} + \text{storage costs} - \text{convenience yield}$$

Relative to spot prices, futures prices are higher when storage costs are higher, and futures prices are lower when the convenience yield is higher. Further, we can say that the shape of the futures price curve depends on available supply (i.e., current inventory of the commodity) along with expected future supply and demand.

Even with these three theories, we are left without a complete theory of commodity futures returns. “Hedging pressure” and “convenience yield” are not observable, and storage costs are not readily disclosed by participant firms.



MODULE QUIZ 30.1

1. The commodity sector that is *least affected* by weather risk is:
 - A. grains.
 - B. precious metals.
 - C. refined energy products.
2. For which of the following commodities is the production and consumption cycle *least affected* by seasonality?
 - A. Hogs.
 - B. Coffee.
 - C. Natural gas.
3. Which of the following factors is *most likely* to distinguish the valuation of a commodity from the valuation of an equity that pays no dividends?
 - A. Holding costs.
 - B. Discount rate.
 - C. Timing of the future sale.
4. A commodity is *most likely* to be physically stored by a(n):
 - A. exchange.
 - B. speculator.
 - C. arbitrageur.
5. A futures market in backwardation will exhibit:
 - A. positive basis and positive calendar spreads.
 - B. negative basis and positive calendar spreads.
 - C. negative basis and negative calendar spreads.
6. Which theory of commodity futures returns is *least likely* to explain why futures markets can be in contango?
 - A. Insurance Theory.
 - B. Theory of Storage.
 - C. Hedging Pressure Hypothesis.

MODULE 30.2: ANALYZING RETURNS AND INDEX CONSTRUCTION



Video covering this content is available online.

LOS 30.g: Describe, calculate, and interpret the components of total return for a fully collateralized commodity futures contract.

An investor who desires long exposure to a commodity price will typically achieve this exposure through a derivative investment in forwards or futures. Some physical commodities cannot be effectively purchased and stored long term, and for others, such as

precious metals, derivative positions may be a more efficient means of gaining long exposure than purchasing the commodities outright and storing them long term.

The return on a derivatives position is not the same as the return on a commodity itself. The total return on a fully collateralized long futures position has three components: collateral return, price return, and roll return.

To take a position in futures, an investor must post collateral. When a futures portfolio is *fully collateralized*, the investor has posted cash or acceptable securities with a value equal to the notional value (price multiplied by contract size) of the futures contracts in the portfolio. If U.S. Treasury bills are deposited as collateral, the **collateral return** or **collateral yield** is simply the holding period yield on the T-bills.

The **price return** or **spot yield** on an investment in commodity futures is the change in spot prices (which can be proxied by futures prices on near-month contracts).

$$\text{price return} = (\text{current price} - \text{previous price}) / \text{previous price}$$

Since commodity derivative contracts expire, an investor who wants to maintain a position over time must close out the expiring futures position and reestablish a new position with a settlement date further in the future. This process is referred to as *rolling over* the position and leads to gains or losses which are termed the **roll return** or **roll yield**. The roll return can be positive if the futures price curve is in backwardation or negative if the futures price curve is in contango.

To hold the value of a long position constant, an investor must buy more contracts if the new longer-dated futures are trading at a lower price (market in backwardation) and buy fewer contracts if the new longer-dated futures are trading at a higher price (market in contango). In any event, the roll return on the contracts traded can be calculated as:

$$\text{roll return} = \frac{\text{price of expiring futures contract} - \text{price of new futures contract}}{\text{price of expiring futures contract}}$$

Roll return is modest over multiple periods but can be a significant portion of total return in a single period.

LOS 30.h: Contrast roll return in markets in contango and markets in backwardation.

Consider a situation where the manager of a portfolio of commodity futures contracts is rolling over July corn futures trading at 397 (cents per bushel) into November corn futures trading at 406. The roll return is:

$$\frac{397 - 406}{397} = -2.27\%$$

With the corn futures market in contango, the roll return is negative.

Now consider a situation where the manager is rolling over July natural gas futures trading at 2.35 (dollars per million cubic feet) into August futures trading at 2.22. In this case the roll return is:

$$\frac{2.35 - 2.22}{2.35} = 5.53\%$$

Suppose we wanted a specific dollar exposure to natural gas, say \$10,000. We would have originally gone long $10,000 / 2.35$, or approximately 4,255 contracts. To maintain the dollar exposure upon rolling over into new contracts, we would have gone long $10,000 / 2.22$, or approximately 4,504 contracts. Hence, when the contract is in backwardation, the roll return is positive and results in a larger number of long contracts upon rolling over.

If natural gas exposure is 8.5% of the manager's portfolio, we can calculate the **net roll return** for the portfolio as $0.085 \times 5.53\% = 0.47\%$.

LOS 30.i: Describe how commodity swaps are used to obtain or modify exposure to commodities.

Swaps can be used to increase or decrease exposure to commodities risk. Swaps are customized instruments created and sold by dealers, who may take on the risk of their swap exposure or hedge their exposure by entering into an offsetting swap contract (in which they have the opposite exposure to the risk factor) or by holding the physical commodity.

Swaps are created for which the payments between the two parties are based on various risk factors such as the excess returns on a commodity, the total return on the commodity, or a measure of price volatility.

In a **total return swap** the swap buyer (the long) will receive periodic payments based on the change in the futures price of a commodity plus the return on the collateral, in return for a series of fixed payments. Each period, the long will receive the total return on holding the commodity times a notional principal amount, net of the payment promised to the short. If the total return is negative, the long makes the promised fixed payment *plus* the negative return percentage on the commodity over the period, times the notional amount.

For example, consider a total return swap on oil with a notional value of \$10 million, in which for two years the long must pay 25 basis points monthly and will receive the total return on West Texas Intermediate (WTI) crude oil. If over the first month the price of WTI increases from 41.50 bbl to 42.10 bbl (+1.45%), the long will receive a net payment of $(0.0145 - 0.0025) \times \$10 \text{ million} = \$120,000$.

If over the second month the price of WTI decreases from 42.10 to 41.20 (−2.14%), the long must make a payment of $(-0.0214 - 0.0025) \times \$10 \text{ million} = \$239,000$ to the short.

Total return swaps are often used by institutions to gain exposure to the price risk of the underlying commodity, avoiding either holding the commodity or managing a long position in futures contracts over time.



PROFESSOR'S NOTE

Up to this point, the various swaps (interest rate, currency, equity) we have considered have comprised two periodic payment streams, with net payments based on the difference between the two. With the following instruments, the buyer may instead make a single payment at the initiation of the swap and then receive periodic payments based on the total returns, excess returns, or price

volatility of a commodity—essentially “buying” exposure to the underlying risk factor.

In an **excess return swap**, a party may make a single payment at the initiation of the swap and then receive periodic payments of any percentage by which the commodity price exceeds some fixed or benchmark value, times the notional value of the swap. In months in which the commodity price does not exceed the fixed value, no payments are made.

In a **basis swap**, the variable payments are based on the difference between the prices of two commodities. Often the two commodities are one that has liquid traded futures available for hedging and the other (the one the swap buyer actually uses in production) with no liquid futures contracts available. Because the price changes of the two commodities are less than perfectly correlated, the difference between them (the basis) changes over time. By combining a hedge using the liquid futures with a basis swap, the swap buyer can hedge the price risk he faces from the input that does not have a liquid futures market.

In a **commodity volatility swap**, the underlying factor is the volatility of the commodity’s price. If the volatility of the commodity’s price is higher than the expected level of volatility specified in the swap, the volatility buyer receives a payment. When actual volatility is lower than the specified level, the volatility seller receives a payment. A similar swap settles based on variance in price levels of a commodity, with a swap buyer receiving a payment if the actual variance exceeds the fixed variance established at the onset of the swap. If the actual variance is lower, the variance seller receives a payment.

LOS 30.j: Describe how the construction of commodity indexes affects index returns.

There are several published commodity indexes. To be most useful, an index should be investable, in that an investor should be able to replicate the index with available liquid futures contracts.

The available commodity indexes differ in the following dimensions:

- Which commodities are included
- The weighting of the commodities in the index
- The method of rolling contracts over as they near expiration
- The method of rebalancing portfolio weights

While no index methodology will consistently outperform another index methodology, differences in methodology do result in returns differences, at least over shorter periods. Over long periods, differences between the mix and weights of constituent commodities in individual indexes will result in differences between returns, as some commodities outperform others.

Indexes may be equal weighted or weighted on some factor, such as the value of global production of an individual commodity or commodity sector. A production value weighted index will have more exposure to energy than to livestock or softs, for example.

With regard to roll methodology, a passive strategy may be to simply roll the expiring futures contracts into the near-month contract each month. A more active strategy would be to maximize roll return by selecting the further-out contracts with the greatest backwardation or smallest contango.

The frequency of rebalancing will also affect commodity index returns. Rebalancing portfolio weights will decrease returns when prices are trending but increase returns when price changes are choppy and mean-reverting. For this reason, price behavior across rebalancing periods will influence returns. If the prices of a commodity are choppy over short horizons but trending on a longer-term basis, frequent rebalancing may capture gains from mean reversion over the shorter periods but give up some of the gains from the trend of the commodity's price over the longer term.

While differences in index construction methodology will lead to differences among index returns over relatively shorter periods, no one methodology is necessarily superior over longer periods. Correlations between returns on different indexes have been relatively high, while correlations between commodity indexes and returns on stocks and bonds have been low.



MODULE QUIZ 30.2

1. Suppose that a commodity market exhibits the following futures curve on July 1, 20X1:

- Spot price: 42.0
- August futures price: 41.5
- October futures price: 40.8
- December futures price: 39.7

An investor establishes a fully collateralized long position on July 1, 20X1, and maintains the position for one year. The futures curve on July 1, 20X2, is identical to the futures curve on July 1, 20X1, and calendar spreads did not change significantly during the year. The investor's total return on the position is *most likely*:

- A. equal to the collateral return.
- B. less than the collateral return.
- C. greater than the collateral return.

2. An investor enters into a swap contract under which the net payment will vary directly with the price of a commodity. This contract is *most accurately* described as a(n):

- A. basis swap.
- B. total return swap.
- C. excess return swap.

KEY CONCEPTS

LOS 30.a

Commodity sectors include energy (crude oil, natural gas, and refined petroleum products); industrial metals (aluminum, nickel, zinc, lead, tin, iron, and copper); grains (wheat, corn, soybeans, and rice); livestock (hogs, sheep, cattle, and poultry); precious metals (gold, silver, and platinum); and softs or cash crops (coffee, sugar, cocoa, and cotton).

Crude oil must be refined into usable products but may be shipped and stored in its natural form. Natural gas may be used in its natural form but must be liquefied to be shipped

overseas.

Industrial and precious metals have demand that is sensitive to business cycles and typically can be stored for long periods.

Production of grains and softs is sensitive to weather. Livestock supply is sensitive to the price of feed grains.

LOS 30.b

The life cycle of commodity sectors includes the time it takes to produce, transport, store, and process the commodities.

- Crude oil production involves drilling a well and extracting and transporting the oil. Oil is typically stored for only a short period before being refined into products that will be transported to consumers.
- Natural gas requires little processing and may be transported to consumers by pipeline.
- Metals are produced by mining and smelting ore, which requires producers to construct large-scale fixed plants and purchase equipment. Most metals can be stored long term.
- Livestock production cycles vary with the size of the animal. Meat can be frozen for shipment and storage.
- Grain production is seasonal, but grains can be stored after harvest. Growing seasons are opposite in the northern and southern hemispheres.
- Softs are produced in warm climates and have production cycles and storage needs that vary by product.

LOS 30.c

In contrast to equities and bonds, which are valued by estimating the present value of their future cash flows, commodities do not produce periodic cash inflows. While the spot price of a commodity may be viewed as the estimated present value of its future selling price, storage costs (i.e., cash outflows) may result in forward prices that are higher than spot prices.

LOS 30.d

Participants in commodity futures markets include hedgers, speculators, arbitrageurs, exchanges, analysts, and regulators.

Informed investors are those who have information about the commodity they trade. Hedgers are informed investors because they produce or use the commodity. Some speculators act as informed investors and attempt to profit from having better information or a better ability to process information. Other speculators profit from providing liquidity to the futures markets.

LOS 30.e

Basis is the difference between the spot price and a futures price for a commodity. Calendar spread is the difference between futures prices for contracts with different expiration dates.

A market is in contango if futures prices are greater than spot prices, or in backwardation if futures prices are less than spot prices. Calendar spreads and basis are negative in contango and positive in backwardation.

LOS 30.f

Insurance Theory states that futures returns compensate contract buyers for providing protection against price risk to futures contract sellers (i.e., the producers). This theory implies that backwardation is a normal condition.

The Hedging Pressure Hypothesis expands on Insurance Theory by including long hedgers as well as short hedgers. This theory suggests futures markets will be in backwardation when short hedgers dominate and in contango when long hedgers dominate.

The Theory of Storage states that spot and futures prices are related through storage costs and convenience yield.

LOS 30.g

The total return on a fully collateralized long futures position consists of collateral return, price return, and roll return. Collateral return is the yield on securities the investor deposits as collateral for the futures position. Price return or spot yield is produced by a change in spot prices. Roll return results from closing out expiring contracts and reestablishing the position in longer-dated contracts.

LOS 30.h

Roll return is positive when a futures market is in backwardation because a long position holder will be buying longer-dated contracts that are priced lower than the expiring contracts. Roll return is negative when a futures market is in contango because the longer-dated contracts are priced higher than the expiring contracts.

LOS 30.i

Investors can use swaps to increase or decrease exposure to commodities. In a total return swap, the variable payments are based on the change in price of a commodity. In an excess return swap, the variable payments are based on the difference between a commodity price and a benchmark value. In a basis swap, the variable payments are based on the difference in prices of two commodities. In a commodity volatility swap, the variable payments are based on the volatility of a commodity price.

LOS 30.j

Returns on a commodity index are affected by how the index is constructed. The index components and weighting method affect which commodities have the greatest influence on the index return. The methodology for rolling over expiring contracts may be passive or active. Frequent rebalancing of portfolio weights may decrease index returns in trending markets or increase index returns in choppy or mean-reverting markets.

Module Quiz 30.1

1. **B** Precious metals mining and smelting are less susceptible to changing weather. Weather is an important factor in grain production with both droughts and flooding affecting crop yields. Oil refineries are concentrated in coastal areas where hurricanes and other extreme weather cause periodic refinery shutdowns. (LOS 30.a)
2. **B** Coffee has a long production cycle but is grown at warm latitudes and harvested throughout the year. Livestock production is strongly influenced by seasonality. Natural gas demand has a seasonal component due to its uses for heating and electricity generation for cooling. (LOS 30.b)
3. **A** While a commodity or a non-dividend-paying equity security can be valued in terms of the present value of its future sales price, a commodity may have holding costs, such as storage, that can result in a forward price that is higher than the spot price. (LOS 30.c)
4. **C** Arbitrageurs may store a physical inventory of a commodity to exploit differences between spot and futures prices relative to the costs of storing the commodity. (LOS 30.d)
5. **A** In backwardation, longer-dated futures contracts are priced lower than shorter-dated contracts or spot prices, resulting in positive basis and calendar spreads. (LOS 30.e)
6. **A** According to Insurance Theory, backwardation is normal because futures contract buyers should earn a positive return for protecting commodity producers (short hedgers) from price risk. The Hedging Pressure Hypothesis and the Theory of Storage can explain either backwardation or contango. (LOS 30.f)

Module Quiz 30.2

1. **C** The price return is zero because the spot price is unchanged over the life of the position. The roll return is positive because the market is in backwardation. Therefore the total return (price return + roll return + collateral return) is greater than the collateral return. (LOS 30.g)
2. **B** In a total return swap, the variable payment is based on the price of a commodity. In an excess return swap, the variable payment is based on the amount by which a commodity price is greater than a benchmark, and the payment is zero if the price is less than the benchmark. The variable payment of a basis swap depends on the difference between two commodity prices. (LOS 30.i)

READING 31

OVERVIEW OF TYPES OF REAL ESTATE INVESTMENT

EXAM FOCUS

Understand the unique features of real estate investments, their valuation models, portfolio characteristics, economic value drivers, and property types, as well as the indexes used to track the performance of real estate investments.

MODULE 31.1: REAL ESTATE FEATURES



Video covering
this content is
available online.

LOS 31.a: Compare important real estate investment features for valuation purposes.

Private real estate investments include direct debt and equity investments in real estate. The real estate industry is composed of firms that develop, service, or operate real estate. Among the kinds of real estate investments are **real estate investment trusts (REITs)**, which own and operate real estate investments and distribute dividends to their shareholders, and **real estate operating companies (REOCs)**, which are taxable companies that own and operate real estate properties with few restrictions.

Classifications

Real estate investment types, in increasing order of risk and return, include the following:

- *Senior debt.* First mortgages, and investment-grade commercial mortgage-backed securities (CMBS).
- *Core.* Stable income-producing properties, diversified public REITs, and sale and leaseback arrangements.
- *Core plus.* A property that is functional and generating income, but has the potential to be improved through Minimal refurbishment, cash flow stabilizing, and tenant repositioning.
- *Value add.* e.g., A property that could be improved through upgrades or repositioning. This category also includes sub-investment-grade CMBS.
- *Opportunistic.* New development/redevelopment, and distressed debt.

One method of classifying real estate is by sensitivity to the business cycle. Defensive investments, such as health care facilities, are relatively immune to fluctuations in the

overall economy. Industrial and office properties are more cyclical investments.

Another way to classify real estate is by assigning the investment to Classes A through D based on the statistical factors that influence real estate values. Class A properties are newer, high-quality construction with top features, while Class D properties are the oldest, with poor-quality construction and no amenities. Class A properties offer high current income but limited capital appreciation potential, while lower property classes offer more opportunity for redevelopment and capital appreciation. Figure 31.1 outlines multifamily property classes.

Figure 31.1: Multifamily Property Classes

Class A	Class B	Class C	Class D
■ Built within the past 10 years or substantially renovated	■ Built within past 20 years, or older property renovated	■ Built within past 30 years, or older property renovated	■ Built over 30 years ago, and located in a less desirable location
■ Top exterior and interior amenities relative to the market	■ Dated interior and exterior amenities relative to market	■ Dated exterior and limited interior amenities relative to market	■ Minimal exterior or interior amenities
■ High-quality construction using highest-quality materials	■ Good-quality construction and only limited deferred maintenance	■ Deferred maintenance on aging construction	■ Low-quality construction, poor condition, and limited remaining usability

Figure 31.2 summarizes the primary modes of real estate investing.

Figure 31.2: Basic Forms of Real Estate Investment

	Debt	Equity
Private	Mortgage debt Construction loans Mezzanine debt	<i>Direct Ownership</i> Sole ownership Joint ventures Limited partnerships
		<i>Indirect Ownership</i> Real estate funds Private REITs
Public	MBS/CMBS/CMOs Covered bonds Mortgage REITs Mortgage ETFs	<i>Publicly Traded Shares</i> Construction Operating Development Public REITs UCITS/mutual funds/ETFs

Source: Reproduced from the Level II CFA Curriculum learning module, "Overview of Types of Real Estate Investment," with permission from CFA Institute.

Financial Metrics

Net operating income (NOI) is the financial metric used most in real estate. Conceptually, it is similar to EBIT in accounting. As shown in Figure 31.3, NOI is the amount of income remaining after subtracting vacancy and collection losses and operating expenses (e.g., insurance, property taxes, utilities, maintenance, and repairs) from **gross potential rental income**. Gross potential rental income reflects current *market* rents, while gross rental income is based on current rents (based on existing signed leases). NOI is calculated before subtracting financing costs and income taxes.

Figure 31.3: Net Operating Income

rental income if fully occupied
+ other income
= gross potential rental income
- vacancy and collection loss
= effective gross income
- operating expense
= NOI

EXAMPLE: Net operating income

Calculate NOI using the following information:

Property type	Office building
Property size	20,000 square meters
Market rent for similar	€250 per square meter
Other income	€75,000
Vacancy and collection loss	5% of potential gross income
Property taxes and insurance	€350,000
Utilities and maintenance	€875,000
Interest expense	€400,000
Income tax rate	40%

Answer:

100% occupancy rent	€5,000,000 [$= 20,000 \text{ m}^2 \times €250/\text{m}^2$]
Other income	75,000
Gross potential income	€5,075,000
Vacancy and collection losses	(253,750) [$= 5,075,000 \times 5\%$]
Operating expenses	(1,225,000) [$= 350,000 + 875,000$]
NOI	€3,596,250

Note that interest expense and income taxes are not considered operating expenses.

Lenders often use the **debt service coverage ratio (DSCR)** and the **loan-to-value ratio (LTV)** to determine the maximum allowable loan amount on a specific property:

$$\text{DSCR} = \text{NOI}_1 / \text{debt service}$$

$$\text{LTV} = \text{loan amount} / \text{appraised value}$$

Debt service (the loan payment) includes both interest and principal. Principal payments reduce the outstanding balance of the loan. (Though in the case of an interest-only loan,

debt service does not reduce the outstanding balance.) Interest rates can be fixed, variable, or have a fixed-rate period followed by a variable-rate period (referred to as an **adjustable-rate loan**).

The ratio (e.g., DSCR, LTV) that indicates the *lowest* amount of allowable debt will determine the maximum loan amount.

EXAMPLE: Maximum loan amount

A real estate lender has agreed to make a 10% interest-only loan on a property that was recently appraised at €1,200,000, provided that the debt service coverage ratio is at least 1.5 and the LTV does not exceed 80%. The property's NOI is €135,000. Calculate the maximum loan amount.

Answer:

Using the LTV, the property will support a loan amount of €1,200,000 appraised value × 80% LTV = €960,000.

Using the DSCR, the property will support a debt service payment of 135,000 NOI / 1.5 = €90,000. The corresponding loan amount is 90,000 payment / 10% interest rate = €900,000.

In this case, the maximum loan amount is €900,000, which is the lower of the two calculated values.

With a loan of €900,000, the LTV is: (900,000 loan amount / 1,200,000 appraised value) = 75%. The DSCR is: (135,000 NOI / 90,000 payment) = 1.5.

When debt is used to finance real estate, equity investors often calculate the **equity dividend rate**. This metric, also sometimes called the **cash-on-cash return**, measures the percentage return on the cash invested by the equity investors:

$$\text{equity dividend rate} = \text{first-year cash flow} / \text{equity}$$

where:

$$\text{first-year cash flow} = \text{NOI} - \text{debt service}$$

The equity dividend rate only takes one period into account. It is not the same as IRR (which measures return over the entire holding period, and includes return from capital appreciation). The equity dividend rate is a pretax metric for the investor.

After-tax return considers the impact of taxes (net of depreciation tax shield) on the investor's return. After-tax return is analogous to ROE. Tax allowance rates for depreciation vary by jurisdiction. The **depreciable base** of a real estate investment excludes the cost of land, which is typically not depreciated.

EXAMPLE: Equity dividend rate and after-tax return

Returning to the previous example, suppose that the property is purchased for the appraised value of €1,200,000, of which €300,000 is the value of the land. The investor's

tax rate is 20%, and the depreciation method for tax purposes is straight line over 30 years.

Calculate the following:

1. Equity dividend rate.
2. After-tax return.

Answer:

1. The €1,200,000 property was financed with €900,000 of debt and €300,000 of equity. First-year cash flow is: (€135,000 NOI – €90,000 debt service payment) = €45,000. Thus, the equity dividend rate is: (€45,000 first-year cash flow / €300,000 equity) = 15%.

2. Depreciation = (€1,200,000 – €300,000) / 30 = €30,000.

Tax = $t \times (\text{NOI} - \text{interest expense} - \text{depreciation}) = 0.20 \times (\text{€135,000} - \text{€90,000} - \text{€30,000}) = \text{€3,000}.$

After-tax return = (€45,000 – €3,000) / €300,000 = 14%.

Principal Risks

The two primary categories of risks affecting real estate investments are as follows:

1. *Economic and competitive factors.* These influence the broad real estate asset class. Such risks include changes in economic activity, changes in demographics, and the cost and availability of capital.
2. *Property-level risks.* These are unique to individual properties. They include obsolescence (requiring extensive remodeling), rezoning, environmental factors (such as earthquakes, floods, wildfires, and energy efficiency mandates), and issues related to managing the property such as maintaining and leasing it.



MODULE QUIZ 31.1

1. A lender is willing to make a 10% interest-only loan on a property, as long as the debt service coverage ratio is at least 1.6 and the loan-to-value ratio does not exceed 80%. The maximum loan amount, if the property has an NOI of \$200,000 and was recently appraised at \$1,500,000, is *closest* to:
 - A. \$1,200,000.
 - B. \$1,250,000.
 - C. \$1,300,000.
2. Which of the following statements about real estate is *least accurate*?
 - A. Class A properties tend to feature higher-quality construction than Class B properties.
 - B. Investments in health care REITs can be considered defensive investments.
 - C. Core-plus property investments tend to exhibit lower risk than core properties do.
3. If residential rental inflation is running at 5%, gross potential rental income is *most likely* to be higher than gross rental income due to:
 - A. vacancy issues.
 - B. collection losses.

C. adjusted rents for new leases.

MODULE 31.2: VALUE DRIVERS AND PROPERTY TYPES



Video covering this content is available online.

LOS 31.b: Explain economic value drivers of real estate investments and their role in a portfolio.

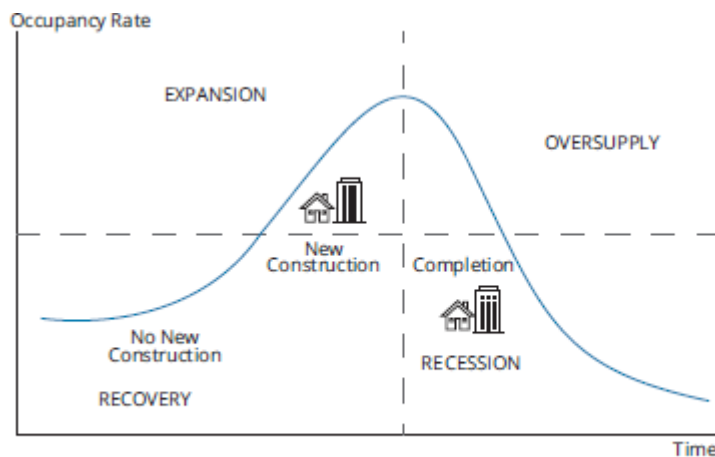
Economic Value Drivers

Common macro factors affecting all types of real estate include GDP growth, wage growth, and job creation. An improving economy initially leads to an increase in rents and a reduction in vacancies, which encourages additional supply, but that supply takes time to emerge because of the long lags in developing real estate.

Job creation and wage growth lead to higher personal income, which results in rising household formation. This establishment of new residences leads to an increase in demand for residential real estate—both rental and owner occupied.

Low home prices relative to incomes, as well as low mortgage rates, tend to increase the demand for owner-occupied housing. This, in turn, spurs new construction. Figure 31.4 shows the typical pattern in the housing market.

Figure 31.4: Real Estate Cycle



Source: Reproduced from the Level II CFA Curriculum learning module, "Overview of Types of Real Estate Investment," with permission from CFA Institute.

The real estate cycle follows several phases:

- The *recovery phase* begins during a business cycle trough. Tight credit conditions and an uncertain business environment discourage new construction, while lower wages put downward pressure on rents. During this phase, interest rates, NOI, and DSC reach a bottom and start rising, while LTV peaks and begins to fall.
- The *expansion phase* coincides with improvement in the business cycle. This leads to wage growth and improved credit conditions, while rents begin increasing—spurring new construction. At the peak of the economic cycle, rising interest rates and property

prices reduce demand for new construction. Interest rates, NOI, and DSC continue to rise, while LTV continues to fall.

- The *oversupply phase* occurs as the economy softens and construction projects in the pipeline are delivered to the market. This contributes to excess supply and decreases prices and occupancy rates. Interest rates, NOI, and DSC peak and begin to fall, while LTV reaches a bottom and begins to rise.
- The *recession phase* occurs as the economy continues to soften, leading to credit tightening and further declines in real estate prices and occupancy rates. Interest rates, NOI, and DSC continue to fall, while LTV increases.

Portfolio Characteristics

Real estate has similarities with and differences from traditional asset classes such as stocks and bonds. Properties generate bond-like fixed lease payments, while also having exposure to equity-like speculative returns from capital appreciation.

Portfolio characteristics of real estate investment include the following:

- *Current income.* Rental income differs from fixed-rate coupons on bonds because leases often include **step-up clauses**. Some leases are variable rate, where the rents are indexed to a market variable such as the Consumer Price Index. Retail properties may have an **overage clause** that results in a higher rent when a tenant's sales exceed a specified level. Some leases require the tenants to bear the operating costs of the property. When the real estate investor's holding period exceeds the lease term, the investor faces **rollover risk** related to finding a new tenant at the end of the lease term.
- *Capital appreciation.* The market value of a property tends to increase over time. Due to the illiquidity and heterogeneity of real estate, a precise measure of capital appreciation is only possible at the time of sale.
- *Inflation hedge.* Real estate can act as an inflation hedge both through rising rents and via property value appreciation. Longer-term leases often use variable rental rates that are indexed to inflation.
- *Diversification.* Low historical price correlations with traditional assets such as stocks and bonds can make real estate an attractive portfolio diversifier.
- *Tax benefits.* Many countries have tax code provisions that provide depreciation deductions, sometimes exceeding the economic loss from deterioration of property value. Also, REITs are exempt from double taxation in many jurisdictions.

LOS 31.c: Discuss the distinctive investment characteristics of commercial property types.

Commercial properties derive their income and value from the economic use of the property, which is influenced by industry-specific factors. Some different types of commercial properties, and their associated value factors and risks, include the following:

- *Office properties.* These may be built to serve the needs of a key tenant (e.g., company headquarters), or to serve many tenants. Traditionally, demand for offices was driven by

business start-ups. However, the work-from-home phenomenon that gained momentum during the COVID-19 pandemic has put a significant dent in demand for office space.

- *Industrial and warehouse properties.* This kind of real estate is sometimes developed for specific clients. Growth in e-commerce has increased demand for warehouse properties. Repurposing these properties to new uses often requires expensive redevelopment.
- *Retail properties.* These include anchor department stores and regional shopping centers. Growth in e-commerce has generally reduced the demand for retail properties, in some cases leading to expensive repurposing. Proximity to local workers and residents is an important attribute for valuing retail properties.
- *Hospitality properties.* These include hotels and other lodging. Their value is influenced by demand for business and leisure travel. Trade shows and conventions are important business drivers for large properties built as convention centers. Business travel is cyclical, which affects room rates and occupancy levels.
- *Mixed use.* These encompass office/retail, residential/retail, and other combinations that derive value from exposure to multiple classes of commercial properties.
- *Other specialty.* This broad classification includes senior housing, self-storage, hospitals, cell towers, data centers, college dormitories, and parking facilities. Each property type has exposure to different risks. The *independent living* category of senior housing—for adults who need only minimal assistance—requires less management. A *continuing care* facility, by contrast, provides seniors more help with daily tasks, which is management intensive. The growing population of seniors is increasing demand for both types of senior housing. Self-storage features short-term rentals, which allows frequent repricing of rents, though occupancy rates can fluctuate.



MODULE QUIZ 31.2

1. Which of the following statements about the real estate cycle is *least accurate*?
 - A. During the expansion phase, interest rates increase, while net operating incomes and debt service coverage ratios decrease.
 - B. When home prices are low relative to household income and mortgage rates are low, investors should be more bullish on residential home builders than on rental housing developers.
 - C. The expansion phase coincides with wage growth and improved credit conditions, which are two factors that spur new construction.
2. Which of the following is *least likely* to represent a portfolio characteristic of a real estate investment?
 - A. Tax benefits.
 - B. Inflation hedging.
 - C. High liquidity.
3. Properties that have more than one tenant type and economic use are called:
 - A. symbiotic centers.
 - B. mixed-use developments.

MODULE 31.3: DUE DILIGENCE, VALUATION, AND INDEXES



Video covering
this content is
available online.

LOS 31.d: Explain the due diligence process and valuation approaches for real estate investments.

Real estate investors normally perform **due diligence** to confirm the facts and conditions that might affect the value of a property. Due diligence includes the following:

- *Market review and outlook.* Investors can aggregate real estate transaction prices (and possibly lease prices) from the internet, based on standardized criteria such as size, location, age, condition, and amenities. Investors can use this data to estimate the value of a specific property. That estimate should take the current phase of the real estate cycle into consideration.
- *Current lease review and future lease outlook.* Here, the investor examines the rental rates and estimates of rent prices going forward, as well as lease expiration dates, vacancy expectations, and the history of rent payments and delinquencies. Forecasts of lease income should also incorporate commission expenses for new leases and the cost of improvements (amortized over the life of the lease) needed to attract future tenants.
- *Financial review.* The investor should confirm operating expenses using evidence such as utility and tax bills, and review maintenance spending to determine if the seller has deferred maintenance to inflate NOI. The investor should ensure that the cash flow statement supports the accrual accounting figures.
- *Documentation review.* A title review (normally required by the mortgage lender) should be conducted to ensure a clean title, free of any liens or ownership disputes. Investors should also check that tax payments are up to date, and whether any restrictions conflict with their intended use of the property.
- *Property inspection and service agreements.* The investor should have a property survey performed to ensure it is free of encroachments or other legal issues. Physical, engineering, and environmental inspections may also be used to identify potential problems with the property. Analyzing service and maintenance agreements (e.g., frequency and cost of repairs) can help an investor identify ongoing problems.

Valuation Approaches

Several different valuation approaches exist for real estate. The most appropriate methodology for a particular property will depend on its characteristics, as well as factors such as market conditions and data availability. An appraiser can arrive at a more reliable estimate of a property's value by combining and averaging approaches.

Cost Approach

The logic behind the **cost approach** is that a buyer is unlikely to pay more for a property than it would cost to purchase similar land and construct a comparable building.

The cost approach involves estimating the market value of the land, estimating the replacement cost of the building, and then adjusting for depreciation and obsolescence. The cost approach is often used for unusual properties, or properties where comparable transactions are limited.

Sales Comparison Approach

The premise of the **sales comparison approach** is that the purchase price should be close to what other buyers have paid for similar properties in the current market. Ideally, the comparable real estate would be identical to the subject property—though in practice, this is rare. Differences may relate to size, age, location, property condition, and market conditions at the time of sale. These mismatches can be addressed by adjusting the sale prices of the comparable properties to reflect their differences from the subject property: the values of comparable transactions are adjusted upward (downward) for undesirable (desirable) differences. In this way, we revalue the comparable as if it were like the subject property.

EXAMPLE: Sales comparison approach

An appraiser has been asked to estimate the value of a warehouse and has collected the following information:

Unit of Comparison	Subject Property	Comparable Transactions		
		1	2	3
Size, in square feet	30,000	40,000	20,000	35,000
Age, in years	5	9	4	5
Physical condition	Average	Good	Average	Poor
Location	Prime	Prime	Secondary	Prime
Sale date, months ago		6	18	12
Sales price		\$9,000,000	\$4,500,000	\$8,000,000

The appraiser's adjustments are based on the following:

- Each adjustment is based on the unadjusted sales price of the comparable.
- Properties depreciate at 2% per annum. Because Comparable 1 is four years older than the subject, an upward adjustment of \$720,000 is made [$\$9,000,000 \times 2\% \times 4$ years].
- *Condition adjustment.* Good: +5%, average: none; poor: -5%. Because Comparable 1 is in better condition than the subject, a downward adjustment of \$450,000 is made [$\$9,000,000 \times 5\%$]. Similarly, an upward adjustment is made for Comparable 3 in the amount of \$400,000 [$\$8,000,000 \times 5\%$].
- *Location adjustment.* Prime: none, secondary: 10%. Because both Comparable 1 and the subject are in a prime location, no adjustment is made.
- Over the past 24 months, sales prices have been appreciating 0.5% per month. Because Comparable 1 was sold six months ago, an upward adjustment of \$270,000 is made [$\$9,000,000 \times 0.5\% \times 6$ months].

Answer:

Once the adjustments are made for all of the comparable transactions, the adjusted sales price per square foot of the comparable transactions are averaged and applied to the subject property as follows:

Adjustments	Subject Property	Comparable Transactions		
		1	2	3
Sales price		\$9,000,000	\$4,500,000	\$8,000,000
Age		+720,000	-90,000	-
Condition		-450,000	-	+400,000
Location		-	+450,000	-
Sale date		+270,000	+405,000	+480,000
Adjusted sales price		\$9,540,000	5,265,000	\$8,880,000
Size in square feet	30,000	40,000	20,000	35,000
Adjusted sales price per SF		\$238.50	\$263.25	\$253.71
Average sales price per SF	\$251.82			
Estimated value	\$7,554,600			

The sales comparison approach is most useful when multiple similar properties have sold recently. This is often the case with single-family homes, but when the market is weak, transactions tend to be fewer. Even in an active market, transactions may be limited for more specialized property types, such as regional malls and hospitals. The sales comparison approach assumes that comparable transactions represent the current market. However, the results of this method can be skewed by a period of irrational pricing, such as a market bubble.

Income Approach

The premise of the **income approach** is that a property's value is equal to the present value of its future cash flows. The rate of return that a buyer requires to invest in the subject property is used to compute this value. Examples of the income approach include the direct capitalization method and the discounted cash flow valuation method.

Under the **direct capitalization method**, value is calculated by capitalizing the first-year NOI of the property using a capitalization rate known as the **going-in cap rate**.

The **capitalization rate**, or **cap rate**, is related to the discount rate, but they are not the same. The discount rate represents the required rate of return and can be thought of as the risk-free rate plus a risk premium.

If NOI and value are expected to grow at a constant rate, the cap rate is lower than the discount rate:

$$\text{cap rate} = \text{discount rate} - \text{growth rate}$$

The cap rate can be applied to produce a valuation:

$$\text{value} = \frac{\text{estimated NOI}}{\text{cap rate}}$$

EXAMPLE: Valuation using the direct capitalization method

Suppose that NOI for an office building is expected to be \$175,000, and an appropriate cap rate has been estimated as 8%. Estimate the value of the property using the direct capitalization method.

Answer:

$$V_0 = \frac{NOI_1}{\text{cap rate}} = \frac{\$175,000}{8\%} = \$2,187,500$$

With the **discounted cash flow method**, value is based on the present value of the property's future cash flows (again, based on NOI) using an appropriate discount rate. A terminal value is used to represent the present value of the property's final cash flows to infinity. Terminal value is estimated as a perpetuity capitalized using the **terminal cap rate**.

Discounted cash flow valuation can be used to estimate value as follows:

$$\text{value} = \sum_{t=1}^n \frac{NOI_t}{(1+r)^t} + \frac{TV_n}{(1+r)^n}$$

where:

$$TV_n = \frac{NOI_{(n+1)}}{\text{terminal cap rate}}$$

r = required rate of return
 n = holding period

EXAMPLE: Valuation using the discounted cash flow method

Suppose that we are considering an investment in an office building with a holding period of three years. It is expected that the NOI for the first year will be \$2 million, and that NOI will grow at 5% per year for the next two years, after which time it will grow at 3% forever. Assume a required rate of return of 8%. Estimate the value of the property using the discounted cash flow method.

Answer:

$$NOI_1 = \$2 \text{ million (given)}$$

$$NOI_2 = 2.00 \times (\$1.05\text{m}) = \$2.10 \text{ million}$$

$$NOI_3 = 2.10 \times (\$1.05\text{m}) = \$2.205 \text{ million}$$

$$NOI_4 = 2.205 \times (\$1.03\text{m}) = \$2.2712 \text{ million}$$

$$\text{Terminal cap rate} = 8\% - 3\% = 5\%$$

$$TV_3 = \$2.2712\text{m} / 0.05 = \$45.424 \text{ million}$$

Using the BA II PLUS financial calculator:

$$CF_0 = 0, CF_1 = 2, CF_2 = 2.10, CF_3 = (2.205 + 45.424) = 47.627$$

Compute NPV → This equals \$41.46 million.

Recall that the cap rate is applied to a property's first-year NOI. An analyst should adjust the subject property's NOI if, due to some temporary situation, it is unrepresentative of similar properties. For example, suppose a property is temporarily experiencing high vacancy during a major renovation. In this case, the first-year NOI should be recalculated as if the renovation was complete. This **stabilized NOI** or **normalized NOI** is then capitalized, and the loss in value from the temporary decline in NOI is subtracted to arrive at the value of the property.

LOS 31.e: Discuss real estate investment indexes, including their construction and potential biases.

Real estate indexes have several uses:

- To track the performance of the real estate asset class
- To serve as a benchmark to evaluate investment managers
- To underpin index-based investment products such as ETFs

Investors should be aware of how the indexes are constructed, as well as their limitations. For example, the observed low correlation of real estate to other asset classes may be partly attributable to the way an index is constructed and the frequency of updates to the index. Private real estate investment indexes are based on property metrics and are usually not investable.

There are two types of real estate investment indexes: appraisal-based indexes and transaction-based indexes.

Appraisal-Based Indexes

Because real estate transactions involving one particular property occur only infrequently, indexes have been developed that instead use appraised values. **Appraisal-based indexes** combine valuations of individual properties to measure the market overall. One popular appraisal-based index in the United States from the National Council of Real Estate Investment Fiduciaries (NCREIF) is the NCREIF Property Index (NPI). Members of the NCREIF (mainly investment managers and pension fund sponsors) submit appraisal data quarterly, and the NCREIF calculates the return as follows:

$$\text{return} = \frac{\text{NOI} - \text{capital expenditures} + (\text{ending market value} - \text{beginning market value})}{\text{beginning market value}}$$

The value of the index is then computed based on value-weighted returns of the individual properties.

Earlier, we defined cap rate as NOI divided by the beginning market value of the property. This is the current yield (or income return) of the property and is one component of the index return equation. The remaining components of the equation contribute the capital

return. To achieve a positive capital return, the property's market value must increase by more than the associated capital expenditures.

Indexes allow investors to compare real estate performance with those of other asset classes. A real estate index can also be used by investors to benchmark real estate returns. However, a major drawback is that the income component of index return does not represent actual cash flows to investors (like REIT distributions do).

The Global Real Estate Fund Index (GREFI) is a capitalization-weighted index, published quarterly, that includes local currency return and incorporates prices going back to 2009. The index combines data from the NCREIF, the European Association for Investors in Non-Listed Real Estate Vehicles (INREV), and the Asian Association for Investors in Non-Listed Real Estate Vehicles (ANREV).

Appraisal Lag

Changes in appraisal-based indexes tend to lag changes in actual prices. Because appraisals use past transaction data as inputs, transactions must occur before appraisal prices can be affected. Appraisals are conducted infrequently; thus, a change in property prices may not be reflected in appraised values until the next quarter or longer. Further, appraisal lag tends to smooth an index—that is, reduce its volatility, much like a moving average does. Because of this smoothing, appraisal lag results in lower calculated correlations with other asset classes.

Appraisal lag can be mitigated by **unsmoothing** the data. To unsmooth an index, a modified autoregressive model can be applied:

$$R^*_t = \alpha R_t + (1 - \alpha) R^*_{t-1}$$

Here, R^* is the appraisal-based return and R is the actual return. The value of α ($0 \leq \alpha \leq 1$) determines the speed with which the appraisal-based return reflects the actual return. A higher α applies more unsmoothing.

Appraisal lag can be avoided by using a transaction-based index.

Transaction-Based Indexes

Transaction-based indexes measure price changes based on actual sales of real estate. Two categories of statistical transaction-based indexes have been developed to address the issue of property sales occurring only infrequently: repeat-sales indexes and hedonic indexes.

A **repeat-sales index** relies on recurring sales of the same property, which helps to distinguish true price changes from differences between properties. With this kind of index, a change in market conditions can only be measured after a property is sold twice. A regression model is used to allocate the changes in index value to each quarter.

A **hedonic index**, by comparison, requires only one sale. A regression model is employed to control for differences in property characteristics such as size, age, and location. Once the model is fitted, it can be used to derive estimates of value for a basket of properties periodically. These data are then used to estimate the returns for that basket.

Public Security Indexes

Bloomberg, FTSE Russell, MSCI, Nikkei, S&P Dow Jones, and Nareit[®], among others, produce indexes covering public securities. Indexes such as CMBX exist for real estate debt securities. Some indexes cover equity REITs only, some REOCs only, and some a combination of the two.



MODULE QUIZ 31.3

1. Which real estate valuation method is *most appropriate* for a 30-year-old, owner-occupied, single-family residence?
 - A. Cost approach.
 - B. Income approach.
 - C. Sales comparison approach.
2. The Royal Oaks office building has annual net operating income (NOI) of \$130,000. A similar office building with NOI of \$200,000 recently sold for \$2,500,000. Using the direct capitalization method, the market value of Royal Oaks is *closest* to:
 - A. \$1,625,000.
 - B. \$2,500,000.
 - C. \$3,846,154.

3. Suppose that a 40-unit apartment building has the following characteristics:

NOI for the next five years	\$600,000
NOI in Year 6	\$700,000
Holding period	5 years
Discount rate	10%
Terminal growth rate	2%

Using the discounted cash flow method, the estimated value of the property is *closest* to:

- A. \$7,707,534.
 - B. \$8,350,729.
 - C. \$9,024,472.
4. The capitalization rate equals the appropriate discount rate:
 - A. less the NOI growth rate.
 - B. plus the NOI growth rate.
 - C. adjusted by a present value factor.
 5. Which of the following statements about the NCREIF Property Index (NPI) is *least accurate*?
The NPI:
 - A. is computed using recent sales transactions.
 - B. is value weighted.
 - C. allows investors to compare real estate performance with that of other asset classes.
 6. Laura Outvie, CFA, enters a contract to purchase a recently renovated apartment building. She is concerned that some of the contractors have not been paid. In performing her due diligence, which of the following procedures should Outvie *most appropriately* conduct to alleviate her concerns?
 - A. Documentation review.
 - B. Financial review.
 - C. Property inspection and service agreement review.
 7. Transaction-based real estate indexes:
 - A. tend to lag appraisal-based indexes.
 - B. tend to be lagged by appraisal-based indexes.

C. appear to have lower correlation with other asset classes than do appraisal-based indexes.

KEY CONCEPTS

LOS 31.a

Real estate investments can be categorized based on whether they represent debt or equity investments, and whether they are publicly traded or private investments. Real estate properties can also be classified based on the degree of sensitivity to the business cycle, or on statistical factors that are proxies for quality.

Net operating income (NOI) is calculated after subtracting vacancy and collection losses and other operating expenses from gross potential rental income, but before subtracting financing costs and income taxes.

Various metrics are used to assess real estate viability and risk:

debt service coverage ratio (DSCR) = $\text{NOI}_1 / \text{debt service}$

loan-to-value (LTV) = $\text{loan amount} / \text{appraised value}$

equity dividend rate = $(\text{NOI} - \text{debt service}) / \text{equity}$

Two categories of risk for real estate investments are (1) economic and competitive factors, and (2) property-level risks.

LOS 31.b

GDP growth, wage growth, and job creation are all value drivers for real estate investments.

The real estate cycle refers to a recurring pattern of four distinct phases that reflect the overall health and direction of the property market. Tight credit conditions in a recessionary trough lead to a halt in new construction as low wages shrink rental income. Expansion of the business cycle leads to improved wages and higher rents. At the peak, high interest rates put a damper on new construction. Oversupply results from softening of demand and delivery of existing projects.

In a portfolio, real estate provides current income, capital appreciation, inflation hedging, and diversification benefits.

LOS 31.c

Types of commercial properties and examples of associated structural changes that may affect value are as follows:

- *Office properties.* Work-from-home arrangements may decrease demand.
- *Industrial and warehouse properties.* Ecommerce has increased demand.
- *Retail properties.* Ecommerce has reduced demand.
- *Hospitality properties.* Demand for business and leisure travel is cyclical.
- *Mixed-use and other specialty properties.* These kinds of real estate are exposed to unique factors.

LOS 31.d

Due diligence for private real estate investments includes the following: market review and outlook; current lease review and future lease outlook; financial review; documentation review; property inspection and service agreement review; examination of property surveys; and review of physical, engineering, and environmental reports.

Valuation approaches for real estate investments include the following:

- *Cost approach*. This involves estimating the market value of the land, estimating the replacement cost of the building, and adjusting the total for depreciation and obsolescence. It is useful for valuing unusual properties or properties where comparable transactions are limited.
- *Sales comparison approach*. The transaction prices of comparable properties are adjusted for differences with the subject property.
- *Income approach (direct capitalization)*. A going-in cap rate is applied to the estimated NOI of the property.
- *Income approach (DCF)*. NOI is estimated for the first n periods. A terminal value at the end of n periods is calculated using the terminal cap rate. These values are discounted and summed to estimate current value.

LOS 31.e

Appraisal-based indexes calculate return as current yield (from NOI) plus price appreciation (adjusted for capital expenditures).

Transaction-based indexes are based on (1) repeat sales, (2) a hedonic model, or (3) public securities. Appraisal-based indexes tend to lag transaction-based indexes; thus, they appear to have lower volatility and lower correlations with other asset classes.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 31.1

1. **A** Using the DSCR, the property will support a debt service payment of: $(\$200,000 \text{ NOI} / 1.6) = \$125,000$. The corresponding loan amount is: $(\$125,000 \text{ payment} / 10\% \text{ interest rate}) = \$1,250,000$. However, using the LTV, the property will only support a loan amount of: $(\$1,500,000 \text{ value} \times 80\% \text{ LTV}) = \$1,200,000$. Thus, the maximum loan that the lender will be willing to extend is \$1,200,000. (LOS 31.a)
2. **C** Core-plus property investments are generally *riskier* than investments in core properties. Class A properties have high-quality construction with the best materials, while Class B properties have good-quality construction with limited deferred maintenance. Health care REITs are classified as defensive investments because demand for medical services is typically independent of the business cycle. (LOS 31.a)
3. **C** When market rents are rising due to positive rental inflation, gross potential rental income (based on current market conditions) will be higher than gross rental

income (based on existing leases). Vacancy and collection losses are deducted from gross potential rental income or gross rental income. (LOS 31.a)

Module Quiz 31.2

1. **A** During the expansion phase, interest rates, NOI, and DSC rise, while LTV decreases. The other statements are accurate. (LOS 31.b)
2. **C** High transaction costs and the relative infrequency of transactions make real estate investments illiquid. Real estate offers current income (rents) and capital appreciation, diversification, and tax benefits, and can be an inflation hedge. (LOS 31.b)
3. **B** Mixed-use developments have more than one tenant type and economic use. (LOS 31.c)

Module Quiz 31.3

1. **C** The sales comparison approach is likely to be the preferred valuation method because data on multiple comparable transactions will normally be available. The cost approach is less applicable here because of the difficulty in estimating depreciation and obsolescence of an older property. The income approach is not appropriate because an owner-occupied property does not generate income. (LOS 31.d)
 2. **A** The cap rate of the comparable transaction is: $(\$200,000 \text{ NOI} / \$2,500,000 \text{ sales price}) = 8\%$. Thus, the estimated value of Royal Oaks is: $(\$130,000 \text{ NOI} / 8\% \text{ cap rate}) = \$1,625,000$. (LOS 31.d)
 3. **A** The terminal value at the end of five years is: $(\$700,000 \text{ Year 6 NOI}) / (10\% \text{ discount rate} - 2\% \text{ growth rate}) = \$8,750,000$. This terminal value is discounted to the present and added to the present value of the NOI over the holding period.
Using our BA II PLUS calculator, we enter the following keystrokes: N = 5; I/Y = 10; PMT = 600,000; FV = 8,750,000; CPT PV = \$7,707,534.
(LOS 31.d)
1. **A** The capitalization rate is the discount rate (required rate of return on equity, r) less the constant growth rate in NOI, g . (Mathematically, $\text{cap rate} = r - g$). (LOS 31.d)
 2. **A** The NPI uses a value-weighted return that is calculated using *appraisal* data. The index allows comparison of the performance of the real estate asset class with that of other asset classes. (LOS 31.e)
 3. **A** Documentation review includes a search of public records for outstanding liens, including those that would have been filed by contractors involved in the recent renovation. (An existing lien can result in legal problems for the purchaser and the lender.) A financial review will not uncover unpaid bills if they were not recorded. A property inspection such as a survey will not identify outstanding liens: rather, a survey confirms the property boundaries and identifies any easements. (LOS 31.d)

4. **B** Appraisal-based indexes tend to lag transaction-based indexes because actual transactions occur before appraisals are performed (appraisals are informed by transaction data). Appraisal-based indexes, not transaction-based indexes, appear to have lower correlations with other asset classes, due to the delay and potential smoothing effect inherent in appraisal-based indexes. (LOS 31.e)

READING 32

INVESTMENTS IN REAL ESTATE THROUGH PUBLICLY TRADED SECURITIES

EXAM FOCUS

For the exam, be able to describe the different types of publicly traded real estate securities and understand the advantages and disadvantages of investing in real estate through publicly traded securities. Understand the three approaches to valuation of real estate property. Familiarize yourself with the process and inputs for the direct capitalization method. Be able to explain the types of REITs, as well as their economic value determinants, investment characteristics, principal risks, and due diligence considerations. Know the various approaches to REIT valuation and be able to calculate the value of a REIT share.

MODULE 32.1: INVESTMENTS IN REAL ESTATE THROUGH PUBLICLY TRADED SECURITIES



Video covering this content is available online.

Publicly Traded Real Estate Investments

Publicly traded real estate securities appeal to small investors seeking to gain exposure to professionally managed real estate. Just like mutual funds, REITs were tax advantaged under the tax codes of major developed markets and became a popular vehicle for investors to gain exposure to real estate asset class.

LOS 32.a: Discuss types of publicly traded real estate securities.

Publicly traded real estate securities can take several forms: **real estate investment trusts (REITs)**, **real estate operating companies (REOCs)**, and residential or commercial **mortgage-backed securities (MBS)**.

We can categorize publicly traded real estate securities into two broad groups: equity and debt.

Equity

Publicly traded real estate equity securities represent ownership stakes in properties.

Equity REITs and REOCs fall into this category.

- *Equity REITs (real estate investment trusts)*. REITs are tax-advantaged companies (trusts) that are for the most part exempt from corporate income tax. Equity REITs are actively managed, own income-producing real estate, and seek to profit by growing cash flows, improving existing properties, and purchasing additional properties. REITs often specialize in a particular kind of property, while still diversifying holdings by geography and other factors. To be tax-advantaged, REITs are generally required to distribute more than 90% of their taxable income to their shareholders.
- *REOCs (real estate operating companies)*. REOCs are not tax advantaged; rather, they are ordinary (i.e., taxable) corporations that own real estate. A business will form as a REOC if it is ineligible to organize as a REIT. For example, the firm may intend to develop and sell real estate rather than generating cash from rental payments, or the firm may be based in a country that does not allow tax-advantaged REITs.

Debt

MBS (mortgage-backed securities) and mortgage REITs fall into this category.

- *Residential or commercial mortgage-backed securities (MBS)*. Residential or commercial mortgage-backed securities are publicly traded asset-backed securitized debt obligations that receive cash flows from an underlying pool of mortgage loans. These loans may be for commercial properties (in the case of CMBS) or on residential properties (in the case of RMBS). Real estate debt securities represent a far larger aggregate market value than do publicly traded real estate equity securities.
- *Mortgage REITs*. Mortgage REITs invest primarily in mortgages, mortgage securities, or loans that are secured by real estate.

REIT Structures

Most REITs are structured as corporations or trusts and are tax-efficient conduits. In most countries, REITs are required to distribute 90% to 100% of their taxable income as dividends to be exempt from taxation. Additionally, REITs have other requirements, such as a minimum 75% investment in real estate and deriving at least 75% of their income from rental or mortgage interest. Ownership rules vary by jurisdiction but tend to disallow family-owned trusts from qualifying as REITs. In the U.S., REITs should have more than 100 owners, and no fewer than five owners can own more than 50% of the REIT shares (5/50 rule).

Advantages

Investments in REITs offer a number of advantages compared with direct investments in physical real estate:

- *Superior liquidity*. Investors in publicly traded real estate securities enjoy far greater liquidity than do investors in physical real estate, because REIT shares trade daily on a stock exchange. The low liquidity of a direct real estate investment stems from the relatively high value of an individual real estate property and the unique nature of each property.
- *Transparency*. Information about current and historical prices and volume is readily available.

- *Exemption from taxation.* As long as certain requirements are met, REITs enjoy favorable taxation, because a major part of REIT distributions is treated as a return of capital and is thus not taxable.
- *Predictable earnings.* The earnings of REITs tend to be relatively consistent over time, because REITs' rental income is fixed by contracts, unlike the income of companies in other industries.
- *Access to premium properties.* Some prestigious properties, such as high-profile shopping malls or other prominent or landmark buildings, are difficult to invest in directly. Shares in REITs that have invested in these properties represent one way to take an ownership stake in these assets.
- *Active professional management.* While a direct investment in properties requires a degree of real estate investment expertise and property management skill, REIT and REOC investments do not. REITs and REOCs employ professional management to control expenses, to maximize rents and occupancy rates, and sometimes to acquire additional properties.
- *Greater potential for diversification.* Because of the high cost of a single property, it is difficult to achieve adequate diversification through direct investments in real estate. Through REITs, however, an investor can diversify across property type and geographical location.

Disadvantages

Most of the disadvantages for REITs as an investment center around the tax code requirement that a REIT distribute most of its earnings as dividends, limiting retained earnings as a means of growth.

- *Limited potential for income growth.* REITs' high rates of income payout limit REITs' ability to generate future growth through reinvestment. This limits future income growth and may dampen the share price of REITs. A REIT may be forced to issue equity at a disadvantageous price to finance its growth, as availability of retained earnings is limited.
- *Lack of flexibility.* The rules that qualify REITs for favorable taxation also have a downside: REITs are restricted in their investment choices. These limits may prevent REITs from being as profitable as they might otherwise be.
- *Lower diversification benefit relative to direct investment in real estate.* REITs tend to have a higher correlation with equities compared to direct real estate investments. So even though REITs allow superior diversification *amongst* property holdings, it offers poorer diversification overall *inside* an investment portfolio.

LOS 32.b: Justify the use of net asset value per share (NAVPS) in valuation of publicly traded real estate securities and estimate NAVPS based on forecasted cash net operating income.

The **net asset value per share (NAVPS)** is the (per-share) amount by which assets exceed liabilities, using current market values rather than accounting book values. NAVPS is generally considered the most appropriate measure of the fundamental value of REITs. If

the market price of a REIT varies from NAVPS, the premium (or discount) reflects an investor's view of the management capabilities, leverage, and governance. REITs with high leverage and trading at a discount to NAVPS may find it difficult to refinance their maturing debt.

NAVPS is superior to book value per share (BVPS) because BVPS relies on the depreciated historical cost of a REIT's asset.

Estimating NAVPS Based on Forecasted Cash Net Operating Income

In the absence of a reliable appraisal, analysts will estimate the value of operating real estate holdings of a REIT by capitalizing the REIT's cash **net operating income** (NOI). This process first requires the calculation of a market-required rate of return, known as the capitalization rate ("cap rate"), based on the prices of comparable recent transactions that have taken place in the market.

$$\text{cap rate} = \text{NOI}_{\text{comps}} / \text{transaction price}_{\text{comps}}$$

NOI is the amount of income remaining after subtracting vacancy and collection losses and operating expenses (e.g., insurance, property taxes, utilities, maintenance, and repairs) from potential gross income. NOI is calculated before subtracting financing costs, depreciation, general and administrative expenses, and income taxes.

Note that in calculating cash NOI, we subtract non-cash rent. Non-cash rent is the difference between the average rent over the term of a lease contract (i.e., straight-line rent) versus the amount of cash rent actually received in a period. Further, in forecasting NOI for the next year, two adjustments are made to the current-period NOI: (1) the impact of acquisitions (if any) for the current year, which are not fully reflected in the current period's NOI, and (2) application of a growth rate.

In the following example, we show how to calculate NAVPS. First, estimated first-year NOI is capitalized using a market cap rate. Next, we add the value of other tangible assets and subtract the value of REIT's liabilities to find total net asset value. Net asset value divided by the number of outstanding shares gives us NAVPS.

Generally, deferred financing expenses, deferred tax assets, and goodwill are excluded from total assets so as to only include "hard economic assets." Similarly, debt and other liabilities are adjusted to their market values if the difference relative to book value is significant.

EXAMPLE: Computing NAVPS

Vinny Cestone, CFA, is undertaking a valuation of the Anyco Shopping Center REIT, Inc. Given the following financial data for Anyco, estimate NAVPS based on forecasted cash net operating income.

Select Anyco Shopping Center REIT, Inc., Financial Information (in \$ millions)

Last 12 months' NOI	\$80
Cash and equivalents	\$20
Accounts receivable	\$15
Total debt	\$250
Other liabilities	\$50
Non-cash rents	\$2
Full-year adjustment for acquisitions	\$1
Land held for future development	\$10
Prepaid/other assets (excluding intangibles)	\$5
Estimate of next 12 months' growth in NOI	1.25%
Cap rate based on recent comparable transactions	8.0%
Shares outstanding	15

Answer:

	Last 12 months' NOI	\$80
–	Non-cash rents ¹	\$2
+	Full-year adjustment for acquisitions ²	<u>\$1</u>
=	Pro forma cash NOI for last 12 months	\$79
+	Next 12 months' growth in NOI (@1.25%/year) ³	<u>\$1</u>
=	Estimated next 12 months' cash NOI	\$80
÷	Cap rate ⁴	<u>8.0%</u>
=	Estimated value of operating real estate ⁵	\$1,000
+	Cash and equivalents ⁶	\$20
+	Land held for future development	\$10
+	Accounts receivable	\$15
+	Prepaid/other assets (excluding intangibles)	<u>\$5</u>
=	Estimated gross asset value	\$1,050
–	Total debt ⁷	\$250
–	Other liabilities	<u>\$50</u>
=	Net asset value	\$750
÷	Shares outstanding	<u>15</u>
=	Net asset value per share ⁸	\$50

Notes:

1. Non-cash rent (difference between average contractual rent and cash rent paid) is removed.
2. NOI is increased to represent full-year rent for properties acquired during the year.
3. Cash NOI is expected to increase by 1.25% over the next year.
4. Cap rate is based on recent transactions for comparable properties.

5. Operating real estate value = expected next 12-month cash NOI / 8% capitalization rate.
6. Add the book value of other assets: cash, accounts receivable, land for future development, prepaid expenses, and so on. Certain intangibles, such as goodwill, deferred financing expenses, and deferred tax assets, if given, are ignored.
7. Debt and other liabilities are subtracted to get to net asset value.
8. NAVPS = NAV/number of outstanding shares.

LOS 32.c: Describe the use of funds from operations (FFO) and adjusted funds from operations (AFFO) in REIT valuation.

Analysts calculate and use two measures: FFO and AFFO.

1. **Funds from operations (FFO).** FFO adjusts reported earnings and is a popular measure of the continuing operating income of a REIT or REOC. FFO is calculated as follows:

$$\begin{array}{rcl}
 & \text{Accounting net earnings} & \\
 + & \text{Depreciation, amortization, impairments, and write-downs.} & \\
 - & \text{Gains from sales of property} & \\
 + & \text{Losses from sales of property} & \\
 \hline
 = & \text{Funds from operations} &
 \end{array}$$

Depreciation is added back under the premise that accounting depreciation often exceeds economic depreciation for real estate. Gains from sales of property are excluded because these are not considered to be part of continuing income.

2. **Adjusted funds from operations (AFFO).** AFFO is an extension of FFO that is intended to be a more useful representation of current economic income. AFFO is also known as **cash available for distribution (CAD)** or **funds available for distribution (FAD)**.

The calculation of AFFO generally involves beginning with FFO and then subtracting non-cash rent and maintenance-type capital expenditures and leasing costs (such as improvement allowances to tenants or capital expenditures for maintenance).

$$\begin{array}{rcl}
 & \text{FFO (funds from operations)} & \\
 - & \text{Non-cash (straight-line) rent adjustment} & \\
 - & \text{Recurring maintenance-type capital expenditures and leasing commissions} & \\
 \hline
 = & \text{AFFO (adjusted funds from operations)} &
 \end{array}$$

Non-cash rent often arises due to straight-lining of rent: non-cash rent is the amount by which the average contractual rent over a lease period exceeds the rent actually paid. A lease calling for increasing rental rates over the lease term will have significant non-cash rent early in the lease term. Capital expenditures related to maintenance, as well as expenses related to leasing the space in properties, are subtracted from FFO because they represent costs that must be expended in order to maintain the value of the properties.

AFFO is considered a better measure of economic income than FFO because AFFO considers the capital expenditures that are required to sustain the property's economic

income. AFFO also provides a better indicator for the sustainability of the REIT's dividend paying capacity. However, FFO is more frequently cited in practice, because AFFO relies more on estimates and is considered more subjective.

Comparison of Different Methods

Net asset value per share: The net asset value method of valuation can be used either to generate an absolute valuation or as part of a relative valuation approach. Note, however, that net asset value is an indication of a REIT's assets to a buyer in the private market, which can be quite different from the value public market investors would attach to the REIT. For this reason, there have historically been significant differences (i.e., premiums or discounts) between NAV estimates and the prices at which REITs actually trade.



PROFESSOR'S NOTE

Relative valuation using NAVPS is essentially comparing NAVPS to the market price of a REIT (or REOC) share. If, in general, the market is trading at a premium to NAVPS, a value investor would select the investments with the lowest premium (everything else held constant).

Advantages of Price Multiples

1. Multiples P/FFO and P/AFFO are globally accepted in evaluating shares.
2. Multiples allow comparisons with other investment alternatives.
3. FFO estimates are widely available from reputed data providers such as Bloomberg.
4. Multiples are evaluated in conjunction with other factors such as growth rates to allow for reconciliation of differences in multiples between different REITs.
5. Because leverage is not explicitly accounted for in FFO and AFFO, analysts need to adjust for leverage differences in relative value analysis.

Drawbacks of Multiples

1. Multiples do not capture the value of non-income-producing real assets (e.g., land held for development, vacant buildings).
2. FFO does not capture the impact of recurring capex-type maintenance expenditures needed to maintain the earning power of properties.
3. One-time gains and accounting charges distort FFO and AFFO and make comparisons between companies difficult.

Discounted cash flow approach: Dividend discount and discounted cash flow models of valuation are appropriate for use with REITs and REOCs, because these two investment structures typically pay dividends and thereby return a high proportion of their income to investors. DDM and DCF are used in private real estate in the same way that they are used to value stocks in general. For dividend discount models, an analyst will typically develop near-term, medium-term, and long-term growth forecasts and then use these values as the basis for two- or three-stage dividend discount models. To build a discounted cash flow model, analysts will generally create intermediate-term cash flow projections plus a terminal value that is developed using historical cash flow multiples.

LOS 32.d: Calculate and interpret the value of a REIT share using the net asset value, relative value (price-to-FFO and price-to-AFFO), and discounted cash flow approaches.

EXAMPLE: Calculating the value of a REIT share

Lucinda Crabtree, CFA, is an asset manager who is interested in diversifying the portfolio she manages through an investment in an office building REIT.

Crabtree wants to value the potential investment using four different approaches as of the end of 2023, as follows:

Approach 1: Net asset value

Approach 2: Price-to-FFO

Approach 3: Price-to-AFFO

Approach 4: Discounted cash flow

Selected REIT Financial Information

	All Amounts in \$ Million
Estimated 12 months' cash net operating income (NOI)	\$80
Last year's actual funds from operations (FFO)	\$70
Cash and equivalents	\$65
Accounts receivable	\$35
Debt and other liabilities	\$400
Non-cash rents	\$5
Recurring maintenance-type capital expenditures	\$15
Shares outstanding	10 million shares
Expected annual dividend next year (2024)	\$5.00
Dividend growth rate in 2025 and 2026	2%
Dividend growth rate (from 2027 into perpetuity)	1%
Assumed cap rate	8%
Office subsector average P/FFO multiple	10×
Office subsector average P/AFFO multiple	14×
Crabtree's applicable cost of equity capital	9%
Risk-free rate	2%

Approach 1: Value of a REIT share using net asset value approach

The value per share for this REIT using net asset value valuation is computed as follows:

Estimated cash NOI	80
Assumed cap rate	<u>8%</u>
Estimated value of operating real estate (80 / 0.08)	1,000
Plus: cash + accounts receivable	100
Less: debt and other liabilities	<u>400</u>
Net asset value	700
Shares outstanding	10
NAV / share	<u>\$70</u>

The REIT share value using the net asset value approach is thus \$70. Note that no adjustment for non-cash rents was required in this case because we began with an estimate of cash NOI.

Approach 2: Value of a REIT share using price-to-FFO (P/FFO) approach

The value per share for this REIT using price-to-FFO valuation is computed as follows:

Funds from operations (FFO)	\$70
Shares outstanding (millions)	10
FFO / share = \$70 million / 10 million shares	<u>\$7</u>

Applying the office subsector average P/FFO multiple of 10× yields a value per share of $\$7 \times 10 = \70 .

The REIT share value using the price-to-FFO approach is thus \$70.

Approach 3: Value of REIT share using price-to-AFFO (P/AFFO) approach

Funds from operations (FFO)	\$70
Subtract: non-cash rents	\$5
Subtract: recurring maintenance-type capital expenditures	<u>\$15</u>
Equals: AFFO	\$50
Shares outstanding (million)	10
AFFO / share = \$50 million / 10 million shares	\$5
Property subsector average P/AFFO multiple	<u>14×</u>

Applying the office subsector average P/AFFO multiple of 14× yields a value per share of $\$5 \times 14 = \70 .

The REIT share value using the price-to-AFFO approach is thus \$70.

Approach 4: Value of REIT share using discounted cash flow approach

	2024	2025	2026	2027
Dividends per share	\$5.00	\$5.10	\$5.20	\$5.25

Present value in 2026 of dividend stream beginning in 2027 = $\$5.25 / (0.09 - 0.01) = \65.63

These dividends are discounted at a rate of 9%.

value of a REIT share

= PV(dividends for Years 1 through n) + PV(terminal value at the end of Year n)

= $PV_{2024 \text{ dividend}} + PV_{2025 \text{ dividend}} + PV_{2026 \text{ dividend}} + PV_{2027 \text{ and later dividends (terminal value)}}$

= $\$5.00 / (1.09) + \$5.10 / (1.09)^2 + \$5.20 / (1.09)^3 + \$65.63 / (1.09)^3$

= \$63.61

The REIT share value using the discounted cash flow approach is thus \$63.61.

Note that the calculated value of a REIT share is likely to vary, sometimes greatly, depending on which of these approaches is used.

LOS 32.e: Explain advantages and disadvantages of investing in real estate through publicly traded securities compared to private vehicles.

We have already identified the advantages and drawbacks of investing in REITs versus private investment (i.e., direct investment) in real estate. We summarize them here.

Advantages of private investment relative to publicly traded real estate securities

1. Direct exposure to the real estate class.
2. Returns dictated by property performance.
3. Tax benefits (accelerated depreciation, timing option for capital gains).
4. Inflation hedge.
5. Illiquidity premium.
6. Control and ability to pursue diverse strategies.
7. Lower correlation with other asset classes (superior diversification).

Drawbacks of private investments relative to publicly traded securities

1. Illiquidity.
2. High fees and expenses.
3. High minimum investment.
4. Low transparency.
5. Fewer regulatory protections for investors.
6. Appraisals lag actual market values.
7. Higher returns derived from the use of leverage, but increases risk.

Advantages of investing in publicly traded securities

1. Tracks real estate asset class fundamentals over the long term.
2. High liquidity.
3. Professional management.
4. Inflation hedge.
5. Tax efficiency (for REITs).

6. Access to a (diversified) pool of assets.
7. Access to special sectors (e.g., data centers).
8. Low minimum investment.
9. Low entry/exit costs.
10. Regulatory protection for investors.
11. High transparency.
12. Limited liability.

Drawbacks of investing in publicly traded securities

1. Higher volatility.
2. Higher correlation with stocks (compared to the correlation of stocks with private real estate).
3. Dividends are taxed at high current income tax rates.
4. Poor governance/agency conflict.
5. Equity markets penalize high leverage.
6. Market prices differ from NAV.
7. REIT structure limits possible activities.
8. Compliance costs may be prohibitive for smaller companies.



MODULE QUIZ 32.1

1. Which of the following *least accurately* identifies one of the principal types of publicly traded real estate securities?
 - A. Commingled real estate fund (CREF).
 - B. Shares of real estate operating companies (REOC).
 - C. Residential and commercial mortgage-backed securities (MBS).
2. Which of the following statements *most accurately* describes one of the advantages of investing in REITs?
 - A. REITs can pass on tax losses to their investors as deductions from their taxable income.
 - B. REITs have lower price and return volatility than a comparable direct investment in properties.
 - C. REITs provide a greater potential for diversification within the asset class.
3. Which of the following statements about the use of net asset value per share (NAVPS) in REIT valuation is *most accurate*?
 - A. NAVPS is the difference between the accounting book values of a real estate company's assets and its liabilities, divided by shares outstanding.
 - B. NAVPS is considered to be a superior measure of the net worth of a REIT's shares, compared with book value per share.
 - C. NAVPS is exactly equal to the intrinsic value of REIT shares.
4. In the process of calculating adjusted funds from operations (AFFO) from funds from operations (FFO), an analyst is *most likely* to:
 - A. add depreciation and amortization.
 - B. subtract non-cash rent.
 - C. add recurring maintenance-type capital expenditures and leasing commissions.
5. Which statement regarding approaches to REIT valuation is *least accurate*?

- A. AFFO includes a number of adjustments to FFO that result in AFFO approximating continuing cash earnings.
- B. P/AFFO is the most frequently used multiple in analyzing the REIT sector.
- C. Dividend discount models are appropriate for valuing REITs because REITs return most of their income to investors.

Use the following information to answer Questions 6 through 9.

Anna Ginzburg, CFA, is using the following information to analyze a potential investment in an industrial building.

Selected REIT Financial Information

	All Amounts in \$ Million
Estimated 12 months' cash net operating income	\$40
Funds from operations (FFO)	\$30
Cash and equivalents	\$30
Accounts receivable	\$20
Debt and other liabilities	\$250
Non-cash rents	\$5
Recurring maintenance-type capital expenditures	\$10
Shares outstanding	10 million shares
Expected annual dividend next year (2024)	\$3.00
Dividend growth rate in 2025 and 2026	4%
Dividend growth rate (from 2027 into perpetuity)	3%
Assumed cap rate	8%
Office subsector average P/FFO multiple	12×
Office subsector average P/AFFO multiple	20×
Ginzburg's cost of equity capital	11%
Risk-free rate	2%

6. The value of Ginzburg's potential investment using a net asset value (NAV) approach is *closest* to:
 - A. \$30.
 - B. \$35.
 - C. \$40.
7. The value of Ginzburg's potential investment using a price-to-FFO approach is *closest* to:
 - A. \$30.
 - B. \$35.
 - C. \$40.
8. The value of Ginzburg's potential investment using a price-to-AFFO approach is *closest* to:
 - A. \$30.
 - B. \$35.
 - C. \$40.
9. The value of Ginzburg's potential investment using a discounted cash flow approach is *closest* to:
 - A. \$30.
 - B. \$35.
 - C. \$40.
10. Compared with other publicly traded shares, REITs are *most likely* to offer relatively low:
 - A. yields.

- B. stability of income and returns.
- C. growth from reinvested operating cash flows.

KEY CONCEPTS

LOS 32.a

The main types of REITs are:

- Equity REITs, which take ownership stakes in income-producing property.
- Mortgage REITs, which invest primarily in mortgages, mortgage securities, or loans that use real estate as collateral.

Advantages of REITs include:

- Superior liquidity.
- Transparency.
- Access to premium properties.
- Active professional management.
- Greater potential for diversification.
- Exemption from taxation.
- Earnings predictability and high yield.

Disadvantages of REITs include:

- Limited potential for income growth.
- Forced equity issuance.

LOS 32.b

Net asset value per share (NAVPS) is the (per-share) amount by which a REIT's assets exceed its liabilities, using current market value of REIT real estate investments rather than accounting or book values. The REIT or REOC portfolio of operating real estate investments can be valued by capitalizing cash net operating income. Adding other assets and subtracting other liabilities yields net asset value.

LOS 32.c

	Accounting net earnings
+	Depreciation, amortization, impairments, and writedowns
–	<u>Gains (losses) from sales of property</u>
=	Funds from operations (FFO)
	FFO (funds from operations)
–	Non-cash (straight-line) rent adjustment
–	<u>Recurring maintenance-type capital expenditures and leasing commissions</u>
=	AFFO (adjusted funds from operations)

LOS 32.d

Price-to-FFO approach:

$$\begin{aligned} & \text{Funds from operations (FFO)} \\ \div & \text{Shares outstanding} \\ = & \text{FFO / share} \\ \times & \text{Sector average P/FFO multiple} \\ = & \text{NAV / share} \end{aligned}$$

Price-to-AFFO approach:

$$\begin{aligned} & \text{Funds from operations (FFO)} \\ - & \text{Non-cash rents} \\ - & \text{Recurring maintenance-type capital expenditures} \\ = & \text{AFFO} \\ \div & \text{Shares outstanding} \\ = & \text{AFFO / share} \\ \times & \text{Property subsector average P/AFFO multiple} \\ = & \text{NAV / share} \end{aligned}$$

Discounted cash flow approach:

$$\begin{aligned} \text{value of a REIT share} = & \text{PV}(\text{dividends for Years 1 through } n) \\ & + \text{PV}(\text{terminal value at the end of Year } n) \end{aligned}$$

LOS 32.e

Relative to publicly traded securities, private investment provides direct exposure to the real estate class, returns dictated by property performance, tax benefits, inflation hedge, illiquidity premium, control, and lower correlation with other asset classes. However, private investments suffer from illiquidity, high fees and expenses, high minimum investment, low transparency, fewer regulatory protections for investors, and higher risk due to leverage.

Publicly traded securities are attractive because they provide high liquidity, professional management, tax efficiency (REITs), diversification, low minimum investment, low entry/exit costs, regulatory protection for investors, and high transparency. However, they suffer from higher volatility, higher correlation with stocks, and agency conflict.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 32.1

1. **A** A commingled real estate fund (CREF) is an example of a private real estate investment, not a publicly traded security. The three principal types of publicly traded real estate securities available globally are real estate investment trusts (REITs), real estate operating companies (REOCs), and residential and commercial mortgage-backed securities (MBS). (LOS 32.a)

2. **C** Compared to direct investment in real estate, a REIT's portfolio investment includes a number of geographically dispersed properties allowing for diversification benefits within the real estate asset class. REITs and REOCs usually cannot pass on tax losses to their investors as deductions from taxable income. Because REIT prices and returns are determined by the stock market, the value of a REIT is more volatile than its appraised net asset value. (LOS 32.a)
3. **B** NAVPS is the difference between a REIT's assets and its liabilities, using current market values instead of accounting book values and dividing by the number of shares outstanding. NAVPS is a superior measure of the net worth of a REIT, compared with book value per share, which is based on historical cost values. NAV is the largest component of the intrinsic value of a REIT; however, other factors, such as the value of non-asset-based income streams, the value added by management, and the value of any contingent liabilities, also contribute to intrinsic value. (LOS 32.b)
4. **B** To calculate AFFO, we begin with FFO and then deduct non-cash rent, maintenance-type capital expenditures, and leasing commissions. (LOS 32.c)
5. **B** FFO has some shortcomings, but because it is the most standardized method of measuring a REIT's earnings, P/FFO is the most commonly used multiple in analyzing REITs. AFFO is used as a convenient proxy for a "cash flow" multiple because AFFO approximates cash earnings. Dividend discount models are appropriate methods for valuing REITs because REITs return a significant portion of their income to their investors and tend to be high-dividend payers. (LOS 32.c)
6. **A** The value per share for this REIT using net asset value valuation is calculated as follows:

Estimated cash NOI	40
Assumed cap rate	<u>8%</u>
Estimated value of operating real estate (40 / 0.08)	500
Plus: cash + accounts receivable	50
Less: debt and other liabilities	<u>250</u>
Net asset value	300
Shares outstanding	10
NAV / share	\$30

The REIT share value using the net asset value approach is \$30. (LOS 32.d)

7. **B** The value per share for this REIT using price-to-FFO valuation is computed as follows:

Funds from operations (FFO)	\$30
Shares outstanding (millions)	10
FFO / share = \$30 million / 10 million shares	\$3

Applying the office subsector average P/FFO multiple of 12× yields a value per share of:

$$\$3 \times 12 = \$36$$

The REIT share value using the price-to-FFO approach is \$36. (LOS 32.d)

8. **A** The value per share for this REIT using a price-to-AFFO valuation is calculated as follows:

Funds from operations (FFO)	\$30
Subtract: non-cash rents	\$5
Subtract: recurring maintenance-type capital expenditures	<u>\$10</u>
Equals: AFFO	\$15
Shares outstanding	10 million
AFFO / share = \$15 million / 10 million shares	\$1.50
Property subsector average P/AFFO multiple	20×

Applying the office subsector average P/AFFO multiple of 20× yields a value per share of $\$1.50 \times 20 = \30 .

The REIT share value using the price-to-AFFO approach is \$30. (LOS 32.d)

9. **C** The value per share for this REIT using a discounted cash flow valuation is calculated as follows:

	2024	2025	2026	2027
Dividends per share	\$3.00	\$3.12	\$3.24	\$3.34
present value in 2026 of dividends stream beginning in 2027 = $\$3.34 / (0.11 - 0.03) = \41.78				
present value of all dividends, when discounted at a rate of 11%				
= $PV_{2024 \text{ dividend}} + PV_{2025 \text{ dividend}} + PV_{2026 \text{ dividend}} + PV_{(\text{terminal value})}$				
= $\$3.00 / (1.11) + \$3.12 / (1.11)^2 + \$3.24 / (1.11)^3 + \$41.78 / (1.11)^3$				
= \$38.15				

The REIT share value using the discounted cash flow approach is \$38.15. (LOS 32.d)

10. **C** When we compare REITs to other kinds of publicly traded shares, REITs offer above-average yields and stable income and returns. Due to their high income-to-payout ratios, REITs have relatively low potential to grow by reinvesting operating cash flows. (LOS 32.e)

READING 33

HEDGE FUND STRATEGIES

EXAM FOCUS

This reading describes some of the most important categories of hedge fund strategies, examines their investment characteristics, and explains how these strategies are implemented. Understand the investment characteristics and implementation approach for six major hedge fund strategy categories. Be able to interpret the model for understanding the risk exposures of each of these strategies.

MODULE 33.1: OVERVIEW OF HEDGE FUND STRATEGIES



Video covering
this content is
available online.

Introduction

Hedge funds represent an important subgroup of alternative investment opportunities.

Key features that distinguish hedge funds from traditional investments include:

- Lower regulatory and legal constraints.
- Flexibility to use short selling and derivatives.
- A larger investment universe.
- Aggressive investment exposures.
- Comparatively free use of leverage.
- Liquidity constraints for investors.
- Lack of transparency.
- Higher cost structures.

The key question is whether the high expense levels of hedge funds are justified by the return and diversification benefits that hedge funds are intended to deliver.

Some asset managers seek out hedge funds as a source of alpha, which tends to be elusive. Others view hedge funds as a way to access top investing talent.

Classifications of Hedge Fund Strategies

Hedge fund strategies are categorized based on the kinds of securities they invest in, the trading approach they use, and the kinds of risk exposures taken.

There are numerous ways to classify hedge funds, but in this topic review, we use the following six strategy categories:

1. *Equity related.* These fund strategies focus on stocks, and hence the primary source of risk is equity risk. There are several subtypes, including long/short equity, dedicated short bias, and equity market neutral.
2. *Event driven.* These strategies relate to corporate actions such as governance activities, mergers, acquisitions, bankruptcies, and other major business events. The main dangers of these strategies are event risks—the possibility that outcomes will not unfold as expected (examples include: the failure of a merger, credit downgrades, or bankruptcy). In the following LOSs, the event-driven hedge fund strategies that we will examine are merger arbitrage and distressed securities.
3. *Relative value.* These hedge fund strategies seek to profit from the price differentials between related securities. For debt securities, the valuation differences being exploited span securities with different credit quality or liquidity. Two relative value strategies that will be considered further are fixed-income arbitrage and convertible bond arbitrage.
4. *Opportunistic.* These strategies employ a top-down approach, often span multiple asset classes, and vary with market conditions. The two opportunistic strategies that will be considered here are global macro and managed futures.
5. *Specialist.* These strategies generally require specialized market expertise or knowledge. Often the risks of these strategies arise due to exposure to specific sectors or unusual securities. The two such specialist strategies we will consider in this topic review are volatility strategies and reinsurance strategies.
6. *Multi-manager.* These strategies use other hedge fund strategies as building blocks, combining different strategies together and rebalancing exposures over time. The two types of multi-manager hedge funds we will consider are multi-strategy funds and funds-of-funds.



MODULE QUIZ 33.1

1. A convertible bond arbitrage strategy is *most likely* to be classified as a(n):
 - A. specialist strategy.
 - B. event-driven strategy.
 - C. relative value strategy.
2. A managed futures hedge fund strategy is *most likely* to be classified as a(n):
 - A. opportunistic strategy.
 - B. specialist strategy.

MODULE 33.2: EQUITY, EVENT-DRIVEN, AND RELATIVE VALUE STRATEGIES



Video covering this content is available online.

LOS 33.b: Discuss investment characteristics, strategy implementation, and role in a portfolio of equity-related hedge fund strategies.

Equity-Related Hedge Fund Strategies

Equity-related hedge fund strategies focus primarily on stock markets, and their risk primarily stems from stocks. Subtypes of the equity-related category include: long/short equity, dedicated short bias, and equity market neutral.

Long/Short Equity

Long/short (L/S) equity hedge funds are straightforward to understand. The fund manager purchases (takes long positions in) stocks that they think will rise in value, and sells (takes short positions in) stocks that they believe will fall in value.

Investment Characteristics

When long/short equity managers combine long and short positions, the resulting portfolio has a beta (i.e., market exposure) equal to the weighted sum of the positive and negative betas of the various long and short positions.

L/S equity hedge funds generally do not seek to *eliminate* market exposure entirely. Rather, L/S funds will typically have a 40% to 60% net long exposure, which is beneficial considering that markets generally trend upward over time.

L/S equity managers typically aspire to provide returns comparable to those of a long-only fund, but with half the amount of standard deviation.

Strategy Implementation

Because successful implementation of a long/short equity strategy requires managers to identify overpriced and underpriced stocks, the majority of L/S equity funds take a sector-specific focus, choosing securities from a particular industry that they are familiar with. They may also use index funds to achieve a desired exposure. L/S funds that are comparatively market neutral may need to use leverage to achieve worthwhile returns.

Role in a Portfolio

The goal of most L/S equity managers is to derive alpha from long and short positions in individual stocks, while also benefitting from a moderate overall long exposure.

When considering an investment in an L/S equity fund, an analyst should weigh whether the investment is worthwhile given the potentially high fees. (Taking a traditional long-only equity position may be a more efficient way to achieve a comparable beta exposure.)

Dedicated Short Selling and Short-Biased

As the name suggests, **dedicated short-selling** funds seek out securities that are overpriced in order to sell them short. **Short-biased** managers use a similar strategy, except that the short position is somewhat offset by a long exposure.

One major challenge of being a short seller is that market prices generally rise over time, which creates a tendency towards negative returns for shorts.

A notable subset of short selling is **activist short selling**, in which the fund manager not only takes a short position in a stock, but also presents research that contends that the stock is overpriced.

Investment Characteristics

Managers using short-biased strategies or dedicated short-selling strategies seek to produce a negative correlation with conventional securities. Compared to other hedge fund strategies, expectations of return for short strategies tend to be lower. Compared to L/S equity hedge funds (which feature offsetting beta exposures) short strategy funds tend to have greater volatility.

Strategy Implementation

Hedge fund managers go “short” a security by borrowing the security and selling it at the current market price. The trade will be profitable if the manager is later able to repurchase the same security at a lower price in order to return it to the lender.

The challenging part of profitably shorting stocks is correctly identifying securities that will lose value. Managers will use a bottom-up approach to identify firms with unprofitable business models, bad management, too much debt, declining market segments, or even deceitful accounting.

A dedicated short seller does not take on any long stock position; rather, they carefully select stocks for a pure short exposure, typically 60% to 120% short. A dedicated short manager who wishes to temper the fund’s market exposure will typically do this by holding cash.

Short-biased managers have a similar strategy; however, they may also take on some long exposure, while remaining net short, often 30% to 60% net short.

For both dedicated short sellers and short-biased managers, relatively little leverage is used.

Role in a Portfolio

The primary goal of dedicated short-selling and short-biased funds is to produce returns that are uncorrelated (or negatively correlated) with the return of conventional portfolio assets. When successful, these negative correlations provide a diversification benefit to a portfolio.

However, this goal of negative correlation comes at a cost: expected returns for short strategies are relatively low.

Equity Market Neutral

Equity market-neutral (EMN) strategies seek to attain a near-zero overall exposure to the stock market. They do this by taking long and short positions in various equities; the weighted betas of these positions should sum to zero. The alpha of EMN strategies is intended to be derived from taking positions in securities that are temporarily mispriced.

Investment Characteristics

The overall goal of EMN funds is to create a portfolio that not only generates alpha, but also is relatively immune to movements in the overall market. Not surprisingly, without beta exposure, the returns of EMN funds tend to be modest. On the other hand, EMN funds can offer significant diversification and low volatility.

Strategy Implementation

EMN fund managers take long positions in particular stocks thought to be temporarily undervalued and short positions in stocks seen as overvalued. When mean reversion eventually occurs, alpha should result.

While some EMN managers (known as discretionary managers) rely on intuition, it is more common for EMN managers (called quantitative managers) to rely on a fixed set of rules to identify trade opportunities.

Because EMN deliberately hedges away market beta, leverage must generally be applied in order to achieve acceptable levels of return.

Popular subtypes of EMN funds include:

- **Pairs trading.** Two stocks with similar characteristics but that are respectively overvalued and undervalued are identified. The co-integrated trading prices of pairs are monitored, and unusual divergence is exploited.
- **Stub trading.** This EMN strategy involves going long and short shares of a subsidiary and its parent company. Generally the positions taken correspond to the percentage of the subsidiary owned by the parent.
- **Multi-class trading.** This strategy entails going long and short relatively mispriced share classes of the same firm—for example, non-voting and voting shares. Profits will arise as the prices of these shares revert to their historical relative valuations.

Aside from investments in stocks, other instruments that can be used to produce a state of zero beta include options, stock index futures, and other derivatives.

Role in a Portfolio

EMN portfolios attempt to produce alpha without taking market beta risk: The composition of EMN funds allows these funds to produce less volatility than funds that rely on beta as a source of return. EMN strategies are particularly successful when markets are volatile and performing poorly.

Event-Driven Strategies

Event-driven hedge fund strategies are those that attempt to profit from accurately predicting the outcome of corporate events, such as bankruptcies, mergers, restructurings, acquisitions, et cetera. To do this, these funds take positions in securities of these corporations or in related derivatives. A **soft-catalyst event-driven** approach is an investment made *before* an event has been announced. A **hard-catalyst event-driven** approach is an investment made *after* a corporate event has been announced; this strategy seeks to take advantage of security prices that have not fully adjusted. Soft-catalyst investing is generally more volatile (and, thus, riskier) than a hard-catalyst approach.

The main risk that impacts event-driven strategies is **event risk**, reflecting the possibility that the outcome of the event will not be the one anticipated. For example, a merger arbitrage hedge fund that anticipates that a particular merger will be successful can suffer a large loss if the merger fails.

In this topic review, we will consider two types of event-driven hedge fund strategies in detail: merger arbitrage and distressed securities.

Merger Arbitrage

Merger arbitrage strategies are investment schemes that attempt to earn a return from the uncertainty that exists in the market during the time between an acquisition being announced and when the acquisition is completed.

Hedge fund managers in the merger arbitrage space profit by correctly anticipating the outcome of various deals. One way to view merger arbitrage is to compare it to writing (selling) insurance on an acquisition. If the acquisition is completed as planned, the hedge fund earns an insurance premium. If the transaction fails, however, the hedge fund stands to lose money—analogous to an insurance company making a payout.

Investment Characteristics

In the case of a merger deal that fails, the price movements that originally occurred when the merger was announced will reverse: the price of the target will fall, and the price of the acquirer will rise. A hedge fund that had taken a position based on the merger succeeding is likely to suffer a significant downside when the deal fails unexpectedly; perhaps on the order of a 40% loss. This kind of potential outcome gives merger arbitrage a significant left-tail risk.

Compared to typical hedge fund strategies, merger arbitrage tends to be more liquid.

Strategy Implementation

In the most common merger arbitrage scenario, the portfolio manager takes positions in the securities of the companies involved, with the expectation of a successful deal. For example, in a stock-for-stock deal, the hedge fund manager will typically purchase the stock

of the target company and *short* the stock of the acquiring company in anticipation of profiting when the deal is completed.

Less commonly, the hedge fund manager may have the opinion that the merger will fail (for example, the deal might be blocked by the government because it would stifle competition). In this scenario, the fund would take the *opposite* positions to those just described.

In order to generate a worthwhile level of return, hedge funds pursuing a merger arbitrage strategy will typically apply 300% to 500% leverage in pursuit of low-double-digit returns.

One specific variety of merger arbitrage involves cross-border mergers and acquisitions (M&A) where two countries and two regulatory authorities are involved. Such mergers are seen as more risky.

Role in a Portfolio

The Sharpe ratios of merger arbitrage strategies tend to be high, as these strategies usually produce relatively steady returns. However, despite these generally steady returns, there is significant left-tail risk.

Distressed Securities

Hedge funds that pursue a **distressed securities** strategy (also called **capital structure arbitrage**) take positions in the securities of companies that are experiencing financial difficulties, including firms that are in bankruptcy or near bankruptcy. Firms may find themselves in this position for a number of reasons, including too much leverage, difficulty competing in their sector, or accounting irregularities. The securities of such a firm will often trade at greatly depressed prices.

Compounding the discounting of the securities of distressed firms is the fact that institutions such as insurance companies and banks are often not permitted to hold non-investment-grade securities. The resultant selling of downgraded distressed securities can create significant pricing inefficiencies and can open up opportunities for hedge funds seeking profit.

When a firm is liquidated, the assets of the company are sold, and the various investors are paid back *sequentially* based on their seniority: senior secured debtholders are paid first, then holders of less-senior secured debt, unsecured debt, convertible debt, preferred stock, and lastly common stock.

As an alternative to liquidation, a firm may instead be able to reorganize, which may involve renegotiating the company's liabilities. Holders of debt may be asked to exchange that debt for new equity or to agree to an extension of the maturity.

Investment Characteristics

Relative to the various event-driven hedge fund strategies, returns of a distressed securities investing strategy tend to be somewhat greater—though generally with larger variability of outcomes.

The **lock-up periods** for investors in event-driven hedge funds tend to be comparatively long (often allowing no redemptions for the first two years). This reflects the extended period of time it can take to value and exit a distressed security investment.

Strategy Implementation

Distressed investing can take different forms. Some managers may make only a passive investment in the distressed securities. Other managers will attempt to acquire the majority of a certain class of security in order to take creditor control during bankruptcy.

Successful distressed securities investing requires a particularly broad range of skills in order to navigate the various legal aspects of the strategy, including the bankruptcy and reorganization proceedings.

While shorting distressed securities is a possibility, the majority of distressed investing takes the form of long investments.

Distressed investing generally makes low use of leverage.

Role in a Portfolio

Distressed securities investing involves moderately high levels of illiquidity due to the nature of the assets being purchased. The returns of distressed securities investing are on-average higher than those of other event-driven strategies, though they are unpredictable and sensitive to declines in the overall market.

LOS 33.d: Discuss investment characteristics, strategy implementation, and role in a portfolio of relative value hedge fund strategies.

Relative Value Hedge Fund Strategies

As the name suggests, **relative value** strategies attempt to exploit valuation differences between securities. The most common securities used in relative value strategies are hybrid convertible debt, though other types of fixed-income securities may be used.

When successful, relative value strategies will earn various premiums over time, including liquidity, credit, and volatility premiums, reflecting differences in liquidity or credit quality between securities. However, during turbulent times, losses can occur.

Two relative value hedge fund strategies that we will consider in detail are fixed-income arbitrage and convertible bond arbitrage.

Fixed-Income Arbitrage

The idea behind **fixed-income arbitrage** strategies is to take advantage of temporary mispricing of fixed-income instruments, by going long undervalued securities, and going short comparatively overvalued securities. Other idiosyncrasies that might be exploited include yield curve kinks or anticipated changes in the shape of the yield curve.

The fixed-income instruments can be of various types, including consumer debt, bank loans, corporate bonds, or sovereign bonds. Hedge fund managers take positions in these

securities in the belief that prices will revert toward their fair values.

Fixed-income arbitrage strategies often make use of significant leverage, in order to produce sufficient levels of return.

Strategy Implementation

Two subtypes of fixed-income arbitrage strategies that we will consider in further detail are (1) yield curve trades and (2) carry trades.

- **Yield curve trades.** In this strategy, the hedge fund manager forms a view of how the shape of the yield curve will evolve over time based on macroeconomic forecasts. The portfolio manager then makes long and short investments in fixed-income instruments in order to profit from the anticipated yield curve steepening or flattening. The portfolio manager will profit when the prices of the long securities rise and the short securities fall. If the positions taken are in securities of different firms, then liquidity, credit, and interest rate risks will be present. For positions in securities of the same issuer, interest rate movements would be the main source of risk.
- **Carry trades.** In a carry trade, the portfolio manager shorts a low-yielding security and goes long a high-yielding security. The source of return here is twofold; first, from the yield differential, and second from the price changes as mean reversion occurs.

Investment Characteristics

Because fixed-income securities tend to be priced efficiently, the amount of profit that can be earned through fixed-income arbitrage is somewhat limited. As a result, substantial leverage is often applied to fixed-income arbitrage strategies. (400% leverage is not uncommon; even 1500% leverage is not unheard of.)

The liquidity of fixed-income arbitrage depends on the particular strategy employed and the kinds of fixed-income instruments used. Strategies involving U.S. Treasuries are very liquid, while strategies that make use of, for example, mortgage-backed securities or foreign instruments will be more difficult to convert to cash.

Role in a Portfolio

Return distributions for a fixed-income arbitrageur tend to be similar to the returns from writing puts. If the trade unfolds as expected, the investor will earn a return from the spread narrowing, plus a return from positive carry. However, if the spread between the two instruments widens unexpectedly, the use of leverage may result in the return to the investor becoming quite negative.

One drawback to fixed-income arbitrage strategies is that their highly leveraged nature can cause modest price volatility to lead to a domino effect of margin calls and deleveraging. For example, the Asian Financial Crisis of 1997 and the Russian Ruble Crisis of 1998 led to the collapse of the renowned hedge fund Long-Term Capital Management.

Convertible Bond Arbitrage

Convertible bonds are fixed-income debt securities that make regular coupon payments but can additionally be exchanged for a prearranged number of common stock shares. The

bond-to-stock conversion is at the bondholder's discretion, though it is only permitted at certain points in the bond's life.

One way to view convertible bonds is as a regular bond, plus a long call option on the corresponding stock.

Analyzing convertible bonds can be complex due to impacts from a number of factors.

The goal of **convertible bond arbitrage** strategies is generally to profit from purchasing the implied volatility of convertible bonds—which is often underpriced. To accomplish this without taking on excess potential for loss, convertible bond arbitrageurs will take other positions in an attempt to hedge the delta and gamma risk of the convertible bond.

Investment Characteristics

Convertible bond arbitrage managers encounter two primary sources of liquidity issues; first, because the strategy requires the manager to short sell the underlying equity, and second, because the fixed-income instruments that are invested in are often complex niche products.

Strategy Implementation

Convertible arbitrage strategies generally exploit the low implied volatilities exhibited by the options within convertible instruments, compared to the historical volatilities of the equities that underlie the option. The challenge is to hedge away the other sources of risk that are embedded in the convertible security including market risk, interest rate risk, and the credit risk of the bond issuer.

When a convertible bond's current conversion price is well above the current stock price, the out-of-the-money call (delta closer to zero) will lead to the convertible bond behaving much like a straight bond, because conversion to stock is unlikely.

Conversely, when a convertible bond's current conversion price is well below the current stock price, the in-the-money call (with delta near one) will cause the convertible bond to behave much like the corresponding stock because conversion is likely.

Significant amounts of leverage are typically applied in implementing convertible bond strategies. For example, a three times long bond exposure and two times short equity exposure might be used. (The smaller short equity exposure stems from delta hedging the short stock exposure according to the delta of the embedded call of the convertible.)

Role in a Portfolio

Convertible arbitrage strategies perform best during periods of normal market conditions: when liquidity is available, when volatility is modest, and when there is a good selection of convertible bonds being issued. Convertible arbitrage may not perform well in periods of illiquidity or acute credit weakness.



MODULE QUIZ 33.2

1. Considering the following equity hedge fund strategies, the strategy that is *most likely* to apply relatively high levels of leverage is a(n):
 - A. equity market neutral strategy.

- B. dedicated short strategy.
 - C. short-biased strategy.
2. An equity-related hedge fund strategy with gross exposures of 80% long and 35% short is *most likely* to be classified as a:
- A. dedicated short strategy.
 - B. short-biased strategy.
 - C. long/short equity strategy.
3. Relative to other hedge fund strategies, equity market neutral strategies are *most likely* to:
- A. exhibit relatively modest returns.
 - B. be vulnerable to periods of market weakness.
 - C. earn return from alpha and beta risk.
4. An investment in distressed securities is *most likely* to be characterized by a:
- A. long bias.
 - B. high level of liquidity.
 - C. large amount of leverage.
5. In a sequential payoff during a liquidation, the security holder that is *most likely* to be paid off first is the holder of:
- A. junior secured debt.
 - B. convertible debt.
 - C. preferred stock.
6. In implementing a convertible arbitrage strategy, a portfolio manager is *most likely* to take a position that is:
- A. long convertible bonds and short equity.
 - B. long straight bonds and short convertible bonds.
 - C. long convertible bonds and short straight bonds.

MODULE 33.3: OPPORTUNISTIC, SPECIALIST, AND MULTI-MANAGER STRATEGIES



Video covering
this content is
available online.

LOS 33.e: Discuss investment characteristics, strategy implementation, and role in a portfolio of opportunistic hedge fund strategies.

Opportunistic Hedge Fund Strategies

Opportunistic hedge fund strategies are a broad class of investing approaches that attempt to extract profits using a wide range of techniques in a broad range of securities. Rather than being focused on individual securities, these strategies take a top-down approach to make macro investments on a global basis across regions, sectors, and asset classes.

The returns of opportunistic hedge fund strategies may be impacted by market cycles, global developments, and international interactions. The risks will depend on the particular strategy and asset classes involved.

Opportunistic hedge fund strategies can be based on a number of broad techniques.

Technical analysis uses past price changes to forecast future price movements. Strategies

based on **fundamentals**, on the other hand, attempt to analyze security prices, markets, sectors, and the relationships between markets, using economic data as the input.

Managers using a **systematic** implementation of their strategies employ computer algorithms and rules to determine which trades to make. Managers using a **discretionary** process instead use their instinct to determine when to trade.

We will consider two opportunistic strategies in detail: global macro and managed futures.

Global Macro Strategies

Managers of **global macro** strategy funds attempt to make correct assessments and forecasts of various global economic variables including inflation, currency exchange rates, yield curves, central bank policies, and the general economic health of different countries.

Global macro managers use a broad range of security types and global asset classes to take positions on these views.

A global macro manager that can identify a global trend early and take a position can profit.

Investment Characteristics

Global macro managers can take positions that are either **directional** (e.g., go long stocks that are anticipated to benefit from expected interest rate hikes, and short stocks that will be disadvantaged), or **thematic** (e.g., buy stocks of firms that will benefit from forthcoming free trade deals).

Unlike some hedge fund strategies, low-volatility mean-reverting markets are not generally favorable for global macro returns.

Because global macro managers take investment positions based on their predictions of the future, there is significant potential for unsuccessful investments when global economies do not behave as expected or if unanticipated risks emerge. As a result, the returns of global macro funds tend to be uneven and volatile.

Strategy Implementation

Global macro strategies are generally based on top-down analysis, beginning with analysis of the global economy, then macro trends within economies, and so on, in order to identify potential opportunities.

Different global macro managers are likely to implement their strategies using very different techniques; for example, technical analysis vs. fundamental analysis, or discretionary implementation versus systematic implementation.

One commonality between global macro funds is that most tend to apply leverage, often representing 600% or 700% of fund assets.

A manager making directional predictions will generally use fundamental information to determine whether a particular asset is undervalued or overvalued on a historical basis. By contrast, a manager using a relative value strategy will seek out securities that are overvalued or undervalued *compared to one another*.

Global macro managers tend to use discretionary approaches more than do managed futures managers.

Role in a Portfolio

When added to a portfolio of traditional assets, a global macro hedge fund can add not only alpha, but also portfolio diversification.

A global macro manager will attempt to anticipate changes before other market participants and then will take a corresponding position and wait for the market to come around. For example, some global macro managers were able to anticipate the United States' subprime mortgage crisis well before the housing bubble began to collapse in 2007.

This contrarian tendency can make an allocation to global macro strategies especially advantageous.

During times of market stress, global macro funds have historically delivered right-tail skewed returns, which is beneficial from a portfolio diversification perspective. However, this behavior cannot always be relied upon, and such diversifying outcomes are not always realized.

Managed Futures

Hedge funds that pursue a **managed futures** strategy take long and short positions in a variety of derivatives contracts including futures, forwards, options on futures, swaps, and sometimes currencies and commodities.

Managed futures strategies can be as simple as trading index futures on a particular sector, or it can involve very exotic contracts such as futures on the weather.

Investment Characteristics

Managed futures funds do not buy and sell assets; rather, they enter into derivatives contracts in order to gain the desired exposures.

Because of the mechanics of futures contracts (requiring only a small amount of upfront collateral), managed futures funds can easily apply great amounts of leverage. Typically, a fund will use perhaps $\frac{1}{8}$ of its capital as collateral on futures contracts. The rest of its capital will be invested in some highly liquid security (such as short-term government bonds) that can also serve as collateral for the futures clearinghouse.

Managed futures funds are extremely liquid, because futures contracts themselves are highly liquid: they trade globally and continuously. Taking long or short futures positions allows a hedge fund manager easy access to exposures across a range of asset classes.

One downside of the popularity of managed futures strategies is that **crowding** has occurred: many market participants pursue the same trades and use similar signals, resulting in a "herding effect" and execution slippage.

Strategy Implementation

There are a number of ways to implement managed futures strategies.

In perhaps the most popular method, **time-series momentum (TSM)** strategies, portfolio managers simply follow the trend: they buy securities that have been rising in price and sell securities that have been trending downward.

Another similar methodology is **cross-sectional momentum (CSM)** strategies, which is carried out within a particular asset class (a *cross section* of assets). Again, the securities rising fastest are purchased while falling securities are shorted.

The high liquidity of futures contracts allows hedge fund managers to pursue a wide selection of trading strategies. Generally, portfolio managers will rely on a signal trigger—most often based on volatility or momentum—to prompt a trade.

In addition to using trade signals, portfolio managers will also have rules for closing a position. Exit methodologies can be based on:

- Price targets.
- Momentum reversal.
- Time.
- Trailing stop-loss.
- Or, some combination of these approaches.

Managed futures investors seek to develop rules and signals that are profitable not only using historical data but also in real-world use. However, the more portfolio managers that use similar signals, the less effective these signals will become.

The size of positions taken are usually based on factors such as correlation and volatility.

Role in a Portfolio

Perhaps the most appealing feature of managed futures is their interaction with other investments. Overall, managed futures have very little correlation with traditional equity and fixed-income assets. The result is that when added to a portfolio, managed futures will generally improve the total risk-adjusted return.

This diversifying characteristic has proven its worth during times of market stress. While other strategies exhibit negative asymmetry during such periods, the right-skewed return distribution of managed futures provides a significant advantage.

LOS 33.f: Discuss investment characteristics, strategy implementation, and role in a portfolio of specialist hedge fund strategies.

Specialist Strategies

Portfolio managers for **specialist hedge fund strategies** use their knowledge of a particular market to pursue niche investment opportunities. The goal of specialist strategies is to generate high risk-adjusted returns that are uncorrelated with those of traditional assets. The risks of such strategies are often unique to the particular niche securities being invested in.

Two such strategies that we will consider in detail are volatility trading and reinsurance/life settlements.

Volatility Trading

Once an esoteric pursuit, **volatility trading** has evolved over recent years to become a recognized investment strategy.

Volatility trading hedge fund managers trade volatility-related assets globally, across countries and across asset classes, in order to exploit perceived differences in volatility prices. The overall goal is to purchase underpriced volatility and sell overpriced volatility.

For example, even though the Tokyo Stock Exchange traditionally has higher volatility than the New York Stock Exchange, implied volatility is usually cheaper in Tokyo than it is in New York. In general, the price of volatility in Asian markets has traditionally been lower than in other regions.

Another volatility trade involves acting as the counterparty to market participants that consistently seek long volatility. Because the negative correlation between stock market returns and equity volatility is high, equity investors seek to buy volatility as a hedge. A hedge fund that is willing to sell volatility will earn an insurance-like premium as compensation for taking on this risk; however, an upturn in volatility can cause such a strategy to unravel in a dramatic fashion.

In the U.S. markets, the most common volatility futures are contracts on the **VIX index**, which tracks the 30-day implied volatility of the S&P 500 index. VIX contracts tend to be mean reverting because high volatility naturally tends to dissipate over time.

Strategy Implementation

Hedge fund managers wishing to pursue volatility trading have several options.

One possibility is to construct various option strategies, such as straddles, calendar spreads, bull spreads, or bear spreads, using basic exchange-traded options.

A second possibility is to make use of over-the-counter (OTC) options, which are customized to meet the portfolio manager's specific needs. A drawback of this method, however, is that it introduces counterparty risk, plus potential liquidity issues.

A more straightforward method of trading volatility is to use futures on the VIX index. An advantage of this approach is that it is a very direct way to express a view on volatility, without the need for hedging. There are some limitations. First, the VIX Index tends to be mean-reverting, which means that trend-following trades tend to be unprofitable. Second, many traders and investors crowd into the VIX futures in order to sell volatility and capture the associated premiums, making it difficult to profit from that strategy.

A fourth method of implementing a volatility trading strategy is to enter into an OTC **volatility swap**, or alternatively a **variance swap**. These derivatives provide a relatively pure exposure to volatility. (Note that the name "swap" here is somewhat misleading. Volatility swaps and variance swaps are actually forward contracts with a payoff based on the difference between observed variance and the expected variance specified in the contract, multiplied by some notional amount.)

Investment Characteristics

The investment characteristics of a volatility trading strategy will vary depending on the securities invested in and the positions taken.

Investors that take a short position in volatility will collect premiums and generally earn stable returns under average market conditions. A long position in volatility will exhibit positive convexity, which can be valuable as a hedge.

The liquidity of a volatility trading strategy will depend on the instruments used. Futures and options based on VIX tend to be extremely liquid, as are exchange-traded volatility options (especially when the tenor is short). On the other hand, OTC contracts are generally less liquid.

The use of futures contracts makes it easy to apply leverage to a volatility trading strategy. The convexity of volatility derivatives means that sometimes large gains can be made from long volatility strategies while taking little risk.

Because of their unique nature, benchmarking of volatility trading strategy performance can be difficult.

Role in a Portfolio

In a portfolio, a long volatility strategy can be a potent diversifier, because stock market volatility is highly negatively correlated with market returns. The disadvantage of maintaining a long volatility position, however, is that a premium must be paid to the volatility seller.

Reinsurance/Life Settlements

In recent years, some hedge funds have sought to take advantage of attractive investment opportunities related to insurance policies. In a typical **life settlement** transaction, an insured person will sell (generally through a broker) their insurance policy to a hedge fund. The hedge fund then will be liable for the premium payments—and will also receive the death benefit upon the passing of the insured.

Individuals sell their life insurance contracts when they feel that they no longer benefit from the agreements. Individuals who purchased life insurance policies may choose to sell their policies to third-party brokers, because those firms will oftentimes pay more for the policy than the issuing insurance company will pay for a surrendered policy.

Catastrophe risk **reinsurance** is another area where hedge funds are increasingly active. **Catastrophe insurance** covers the policy holder against earthquakes, tornadoes, hurricanes, floods, and the like. In order to diversify and decrease risk, insurance companies in their normal course of business will sell off some of their risk to reinsurance companies, who may then resell these risks to hedge funds. Reinsurance can be a rewarding investment for a hedge fund if sufficient diversity can be obtained (i.e., the risks should vary by geography and types of insurance), if the insurance company provides sufficient loan loss reserves, and if the policy premiums are adequate. When considering an investment in catastrophic insurance, a hedge fund evaluates both typical and worst-case outcomes against the insurance premiums to be received. Geographic diversity is

important, because a specific catastrophic event will tend to affect only a particular part of the world.

Investment Characteristics

Strategies that involve investments in insurance contracts are illiquid, because insurance policies are somewhat difficult to sell after initiation.

Strategy Implementation

The term “life settlement” refers to a secondary market transaction on an insurance policy. A hedge fund that invests in life settlements will analyze various pools of life insurance contracts that brokers offer, and invest in the ones that they expect to produce an attractive return. After investing, the hedge fund then becomes the beneficiary of these contracts. The investment is successful if the present value of the future insurance payout exceeds the present value of the payments made by the hedge fund.

In selecting which insurance policies to invest in, a hedge fund will seek out policies with the following characteristics:

- The purchase price of the policy is low.
- The ongoing premium payments are low.
- The insured person is likely to die relatively soon.

One major prerequisite to profiting from life settlements is an accurate alternative estimate of life expectancies. Appraising a life settlement requires a significant amount of skill and knowledge and requires comparing individual policyholders’ outlooks to actuarial averages.

Role in a Portfolio

An appealing feature of insurance investments in a portfolio is that the risk inherent in these strategies is almost entirely uncorrelated with market risks and business cycles. For example, floods and earthquakes usually have little correlation to financial markets. Thus, hedge funds that invest in such instruments may be able to increase portfolio alpha, while simultaneously adding return diversification.

LOS 33.g: Discuss investment characteristics, strategy implementation, and role in a portfolio of multi-manager hedge fund strategies.

Multi-Manager Hedge Fund Strategies

Up to this point in the topic review, we have been considering various individual hedge fund strategies. In reality, most investors put money into not just a single hedge fund category, but into a diverse set of strategies.

The notion behind multi-manager hedge funds is to assemble in a deliberate way a portfolio of diverse hedge fund strategies and to adjust the holdings strategically over time.

The most common styles of multi-manager hedge funds are funds-of-funds and multi-strategy funds.

Fund-Of-Funds

A **fund-of-funds (FoF)** takes capital from various individual investors and invests in a number of different hedge funds, generally each pursuing a different strategy.

FoF can provide investors with a number of benefits:

- Diversification across many hedge fund strategies.
- Expertise in individual manager selection.
- Strategic allocation and style allocation.
- Due diligence.
- Occasional value-added tactical decisions.
- Currency hedging.
- Leverage at the portfolio level.
- Better liquidity relative to individual hedge funds.
- Access to certain closed hedge funds.
- Economies of scale in fund monitoring.
- Research expertise.
- Potentially valuable concessions from the underlying funds.

FoF also have the following disadvantages:

- A second layer of fees for the investor.
- Lack of transparency into individual hedge funds.
- No netting of performance fees.
- Additional principal-agent issues.

Investment Characteristics

Individual hedge funds have traditionally employed a “**2 and 20**” fee structure, indicating management fees of 2%, plus performance incentive fees of 20%. On top of these individual fund fees, funds-of-funds have historically added a 1% management fee, plus a further 20% incentive fee on the *total* FoF portfolio. (Though over time, these FoF fees have become negotiable and generally smaller.)

One important benefit that FoF offer is making an investment in hedge funds practical for smaller investors, such as small institutions or moderately wealthy individuals. Individual hedge funds almost always require a significant initial investment; typically \$1 million. Achieving a diversified exposure to a number of individual funds poses a difficulty for smaller investors, because an investment in 20 such hedge funds might require a cumulative minimum investment on the order of \$20 million. Furthermore, performing due diligence on 20 different hedge funds would require far more resources than most individual investors could muster. (Not to mention complications such as tax reporting on 20 investments.) Using a FoF, on the other hand, a smaller investor can typically access a diversified hedge fund exposure with an investment as small as \$100,000. In this way, FoF can serve as an entry point into hedge fund investing.

Aside from providing a straightforward path into hedge funds, FoF provide a number of other advantages. For example, FoF may provide access to high-profile managers whose funds are otherwise closed to new investors. Also, the larger size of an FoF may allow the fund to obtain valuable concessions from the underlying funds' management.

Liquidity can be a challenge for portfolio managers of funds-of-funds. Typically, a FoF will require a one-year initial **lock-up** for investors, and then will allow somewhat greater liquidity afterwards (e.g., monthly or quarterly). However, the underlying funds may have stricter limits on liquidity, potentially putting an FoF manager in a squeeze.

Another drawback of funds-of-funds relates to netting risk: investors could be required to make substantial incentive payments to a small number of successful underlying funds, even if the overall performance of an FoF is poor.

Strategy Implementation

A fund-of-funds strategy is normally implemented as follows:

1. Use fund databases plus personal introductions to become familiar with hedge funds available for investment.
2. Choose an appropriate strategic allocation to different hedge fund strategies.
3. Initiate the manager selection process, applying both top-down and bottom-up techniques.
4. For each hedge fund strategy, consider a number of candidates following that strategy.
5. Interview the candidate hedge fund managers.
6. Review relevant materials such as audit reports.
7. Examine the funds' personnel, operational processes, and risk management.
8. Negotiate with the individual fund managers for lower fees, improved liquidity, or other terms.
9. After the various funds are approved and included in the FoF, an ongoing monitoring process begins, intended to detect major personnel changes, style drift, etc.

The FoF's strategic allocation determines the percentage of total capital that will be invested in each hedge fund style. In addition to this strategic allocation, there may also be a tactical allocation, whereby the FoF manager will at various times underweight or overweight the various hedge fund strategies to reflect the FoF manager's perception of a changing market environment.

Role in a Portfolio

When an FoF manager takes a number of relatively uncorrelated hedge funds and combines them together in the same portfolio, the resulting FoF should produce a number of advantages: greater diversification, steady returns, less concentrated exposure to risks, less volatility, and less exposure to the downside risk of any individual fund manager.

Multi-Strategy Hedge Funds

Like an FoF, **multi-strategy** hedge funds are funds that hold a number of other hedge funds where these various funds are pursuing diverse strategies. Also similar to an FoF, multi-

strategy hedge funds are intended to use this diversification of strategy to produce steady, low-volatility returns.

Unlike an FoF, the sub-funds in multi-strategy hedge funds are run by the *same* organization, rather than being managed by different hedge fund firms.

Investment Characteristics

The diversification within multi-strategy funds is intended to produce steady returns and low volatility.

Multi-strategy funds share some investment characteristics with funds-of-funds, but there are also significant differences. For example, the **operational risks** of a multi-strategy fund are not well-diversified as they are with an FoF, because all of the operational processes of multi-strategy funds are performed under the same roof.

Furthermore, the diversity of strategies represented by the different funds in multi-strategy funds are often somewhat limited, because the managers employed by a specific multi-strategy fund tend to have similar investment viewpoints and methods.

A major advantage of multi-strategy funds over an FoF is the speed and relative ease with which **tactical allocations** can be made. Because each of the multi-strategy funds are managed in-house, it is relatively easy for the multi-strategy manager to reallocate capital from one strategy to another. The high internal transparency and fast response time makes tactical reallocations of multi-strategy funds practical, which could explain why multi-strategy funds have historically been perceived as superior to funds-of-funds for protecting investments.

Investor fees for a multi-strategy fund are often more attractive than those of an FoF. While FoF investors are subject to **netting risk** (where hefty performance fees can be paid to some successful sub-fund managers, despite overall poor FoF performance), multi-strategy funds are more likely to absorb this netting risk internally. In this arrangement, the investor only pays an incentive fee on the *total* fund performance. (Though some multi-strategy funds do use an FoF-like “pass-through” fee model, which will expose investors to netting risk.)

Like an FoF, multi-strategy funds generally limit investor liquidity using redemption periods and initial lock-ups. Multi-strategy funds often additionally enforce limits on the amount of redemption each quarter.

Strategy Implementation

Multi-strategy funds carry out their approach by making investments in a number of varying hedge fund strategies.

One key advantage of multi-strategy funds is their ability to make tactical reallocations, in addition to the funds’ strategic allocation. Furthermore, the multi-strategy fund’s internal teams are likely to be well informed about why and when capital and leverage should be reallocated, versus a FoF manager for whom the various funds are more opaque.

Risk management can be more effective with a multi-strategy fund because (unlike FoF managers) multi-strategy managers should have a solid understanding of correlations and common risks between the various funds.

Multi-strategy funds also enjoy efficiencies that come from several hedge fund teams sharing the same administrative resources.

Multi-strategy funds often make greater use of leverage than does the average FoF. Normally, leverage in multi-strategy funds does not pose much of a hazard; however, during periods of market stress, small sources of danger can become significant left-tail risks that threaten the survival of the fund. (This kind of scenario led to the demise of Ritchie Capital in 2005 and Amaranth Advisors in 2006.)

Multi-strategy funds generally have more varied performance than does an FoF.

Role in a Portfolio

Multi-strategy funds are intended to improve an investment portfolio by adding diversification and steady, low-volatility returns.

Historically, multi-strategy funds have generally performed better than have funds-of-funds, due to a superior fee structure and greater ability to execute on tactical asset allocation. However, the leveraged nature of multi-strategy funds can sometimes lead to a left-tail blow-up during times of stress.



MODULE QUIZ 33.3

1. Considering global macro strategies and managed futures strategies, it would be *most accurate* to state that:
 - A. managed futures strategies use more discretionary approaches.
 - B. global macro strategies use more systematic approaches.
 - C. both strategies tend to be highly liquid and use high leverage.
2. During periods of market stress:
 - A. managed futures and global macro both exhibit right-tail skewness.
 - B. managed futures strategies exhibit left-tail skewness.
 - C. global macro strategies exhibit left-tail skewness.
3. Considering the correlation between equity volatility and equity market returns, the two measures are *most likely* to be:
 - A. highly positively correlated.
 - B. predominantly uncorrelated.
 - C. highly negatively correlated.
4. A hedge fund is *most likely* to purchase a pool of life insurance policies that has high:
 - A. surrender value.
 - B. ongoing premium payments.
 - C. likelihood of the insured person dying soon.
5. Compared to a multi-strategy fund, a fund-of-funds is *most likely* to offer the investor a more:
 - A. effective tactical asset allocation.
 - B. attractive fee structure.
 - C. diverse strategy mix.
6. Compared to a multi-strategy fund, a fund-of-funds is *most likely* to present an investor with higher:
 - A. transparency.
 - B. netting risks.

MODULE 33.4: FACTOR MODELS AND PORTFOLIO IMPACT OF HEDGE FUNDS



Video covering
this content is
available online.

LOS 33.h: Describe how factor models may be used to understand hedge fund risk exposures.

Factor Models

Analysis of Hedge Fund Strategies

In this LOS, we will consider the use of a **conditional linear factor model** to quantify the risk exposures of various hedge fund strategies.

By *conditional*, we mean a model that takes into account that a fund may behave one way during normal market conditions, but perform differently during a period of market turbulence (such as the economic crisis of 2007–2009).

By determining a fund's actual risk exposures, events such as the widespread closure of hedge funds that happened after the global financial crisis are less likely.

Conditional Factor Risk Model

Suppose we use the following conditional linear factor model to explain the return of hedge fund i in period t :

$$\begin{aligned} (\text{Return on HF})_{it} = & \alpha_i + \beta_{i,1}(\text{Factor 1})_t + \beta_{i,2}(\text{Factor 2})_t + \dots + \beta_{i,K}(\text{Factor } K)_t + \\ & D_t\beta_{i,1}(\text{Factor 1})_t + D_t\beta_{i,2}(\text{Factor 2})_t + \dots + D_t\beta_{i,K}(\text{Factor } K)_t \\ & + (\text{error})_{it} \end{aligned}$$

where:

α_i = intercept for hedge fund i

$\beta_{i,K}(\text{Factor } K)_t$ = exposure during *normal* periods to risk factor K

D_t = dummy variable that equals zero during normal periods, and one during a financial crisis

$D_t\beta_{i,K}(\text{Factor } K)_t$ = *incremental* exposure to risk factor K during financial crisis periods

$(\text{error})_{it}$ = random error with zero mean

Any returns not explained by the model's risk factors would be attributed to either omitted risk factors, alpha (i.e., hedge fund manager skill), or randomness (error).

Hasanhodzic and Lo (2007) used the following six factors:

1. *Equity risk (SNP500)*: S&P 500 total return index.
2. *Interest rate risk (BOND)*: Bloomberg Barclays Corporate AA Intermediate Bond Index.
3. *Currency risk (USD)*: U.S. Dollar Index.
4. *Commodity risk (CMDTY)*: Goldman Sachs Commodity Index (GSCI) total return.
5. *Credit risk (CREDIT)*: Spread between Moody's Baa and Aaa corporate bond yields.
6. *Volatility risk (VIX)*: CBOE Volatility Index (VIX).

A **stepwise regression** process is useful for creating linear conditional factor models that avoid multicollinearity problems, because it avoids the use of highly correlated risk factors. Use of a stepwise regression process resulted in the BOND and CMDTY factors being dropped due to multicollinearity issues. (CREDIT and SNP500 respectively produced higher adjusted R^2 .)

This left the following four factors for measuring risk exposures:

1. Equity risk (SNP500).
2. Currency risk (USD).
3. Credit risk (CREDIT).
4. Volatility risk (VIX).

Each hedge fund strategy has different exposures to these various risk factors. These risk factors stem from taking long or short positions in financial instruments that are exposed to these risks.

For example, arbitrage strategies are often exposed to credit spread risk and market volatility risk. Event-driven strategies and L/S equity strategies generally have significant exposure to equity (market beta) risk.



PROFESSOR'S NOTE

The curriculum variously refers to this conditional factor risk model as a “factor model,” “linear factor model,” “conditional risk model,” etc. These terms are all intended to refer to the same idea.

LOS 33.i: Evaluate the impact of an allocation to a hedge fund strategy in a traditional investment portfolio.

We will assume a 20% allocation to a hedge fund strategy from a conventional portfolio comprised of 60% stock and 40% bond. After the allocation, the new weights will be 20% hedge fund, 48% stock, and 32% bond.

Portfolio Contribution of Hedge Fund Strategies

Performance Contribution to a 60/40 Portfolio

For most strategies, when added to a conventional portfolio:

- Total portfolio standard deviation decreases.
- Sharpe ratio increases.
- Sortino ratio increases.
- Maximum drawdown decreases (in approximately one-third of portfolios).

The interpretation of these results is that hedge fund strategies generally increase risk-adjusted return and provide diversification to a traditional portfolio of stocks and bonds.

Risk-Adjusted Performance

The **Sharpe ratio** is one risk-adjusted measure of performance. The risk metric used to calculate the Sharpe ratio is standard deviation—with the result that both downside *and* upside deviations result in a lower Sharpe ratio.

The **Sortino ratio** is a similar risk-adjusted measure of performance; however, only *downside* deviations are considered to reflect risk: risk is measured as variability *below* a predefined level of return. Because of the left-tail risk present in many hedge fund strategies, the Sortino ratio is generally seen as a superior measure of the risk-adjusted performance of hedge funds.

When 20% allocations to various hedge fund strategies are added to the traditional stock/bond portfolio, particularly high Sharpe ratios are achieved from allocations to these strategies:

- Systematic futures.
- Distressed securities.
- Fixed-income arbitrage.
- Global macro.
- Equity market neutral.

The highest Sortino ratios were attained via allocations to the following hedge fund strategies:

- Equity market neutral.
- Systematic futures.
- Long/short equity.
- Event driven.

Allocations to the following strategies were found to be effective in generating superior risk-adjusted performance, based on comparatively high Sharpe and Sortino ratios:

- Systematic futures.
- Equity market neutral.
- Global macro.
- Event-driven hedge fund strategies.

On the other hand, it was observed that the following fund strategies do not significantly enhance risk-adjusted performance:

- Fund-of-funds.
- Multi-strategy.

While all of the hedge fund strategies expose the overall portfolio to various kinds of additional portfolio risks, allocating a portion of a stock/bond portfolio to hedge funds generally reduces risk and increases returns.

Risk Metrics

One key reason for investors to allocate a portion of their portfolio to hedge fund strategies is to reduce risk.

Standard Deviation

Perhaps not surprisingly, it was found that the following strategies resulted in the lowest standard deviations of returns for the overall portfolio:

- Dedicated short-biased.
- Bear market neutral.

These funds also produced notably low standard deviations:

- Systematic futures.
- FoF: macro/systematic.
- Equity market neutral.

The risk-reduction ability of these strategies has been found to be substantial. Not surprisingly, they are also some of the strategies that most improve risk-adjusted returns.

Funds that were found to have little positive impact on reducing standard deviations of the overall portfolio include:

- Event-driven: distressed securities.
- Relative value: convertible arbitrage.

An explanation for event-driven: distressed securities' lack of ability to reduce standard deviation is that these funds tend to take long positions in securities, and outcomes are either mild successes or grand failures.

Relative value: convertible arbitrage funds do little to improve standard deviation, likely because this strategy's leveraged nature becomes a liability during times of market volatility.

Drawdown

Drawdown is defined as the peak-to-trough decline for a portfolio, generally quoted as the percentage drop between a point of high portfolio value and the subsequent trough. The **high-water mark** refers to the maximum value the portfolio has *ever* reached.

Drawdown is useful for quantifying the risk and past performance of various investment strategies, and for comparing one strategy's risk to another.

The hedge fund strategies that produced the smallest maximum drawdowns are various opportunistic strategies:

- Global macro.
- Systematic futures.
- Merger arbitrage.
- Equity market neutral.

Use of the conditional risk model can show that these strategies perform relatively well during periods of market crisis because they have minimal exposure to credit risk or equity. Furthermore, these strategies benefit from their liquid nature. Those properties make opportunistic strategies useful diversifiers for traditional assets.

Other hedge fund strategies did little to mitigate the traditional portfolio's maximum drawdown:

- Long/short equity.
- Event-driven: distressed securities.
- Relative value: convertible arbitrage.

These results are somewhat expected. Using the conditional risk model, we can demonstrate that these particular strategies have significant exposure to equity risk, and furthermore during crisis periods they have significant exposure to credit risk as well.



MODULE QUIZ 33.4

1. Conditional linear factor models used to understand hedge fund risk exposures are *most likely* to use factors that include:
 - A. liquidity risk, operational risk, valuation risk, and systemic risk.
 - B. interest rate risk, commodity risk, margining risk, and concentration risk.
 - C. equity risk, credit risk, currency risk, and volatility risk.
2. Adding a 20% allocation of a hedge fund strategy to a traditional 60%/40% portfolio is *most likely* to increase the total portfolio's:
 - A. standard deviation.
 - B. maximum drawdown.
 - C. Sortino ratio.
3. The risk-adjusted performance of a traditional 60%/40% portfolio is *most likely* to be improved by adding an allocation to a hedge fund using the strategy of:
 - A. equity market neutral.
 - B. fund-of-funds.
 - C. multi-strategy.

KEY CONCEPTS

LOS 33.a

Hedge fund strategies are classified according to the instruments they invest in, the philosophy followed, and the kinds of risk exposures taken.

This reading classifies hedge fund strategies into the following six categories:

1. Equity related.
2. Event driven.
3. Relative value.
4. Opportunistic.
5. Specialist.
6. Multi-manager.

LOS 33.b

Long/short equity. This strategy generates alpha via careful stock picking. L/S funds are typically liquid, and generally net long. Equity L/S managers aspire to the returns of a long-only approach, but with lower standard deviations. The more market-neutral the approach, the more leverage is likely to be applied.

Dedicated short-selling and short-biased strategies. These two strategies have negative correlations to traditional assets and modest return goals. The focus is on stock picking, using minimal leverage. Dedicated short strategies are generally 60% to 120% short at all times. Short-biased strategies are typically 30% to 60% net short: the short beta exposure is moderated with some long exposure (and cash).

Equity market-neutral (EMN). These strategies attempt to profit from short-term mispricing between securities. Beta risk is minimal, meaning that EMN strategies can perform well in periods of market weakness. Most managers are quantitative (vs. discretionary). High leverage is usually used.

LOS 33.c

Merger arbitrage. This strategy attempts to profit by taking positions on a corporate takeover. Merger arbitrage returns are usually insurance-like, with a high Sharpe ratio. However, left-tail risk is present: negative returns can occur if a merger deal unexpectedly fails. Some leverage is usually applied to generate meaningful returns. Merger arbitrage is a relatively liquid strategy.

Distressed securities. These strategies seek out mispriced securities of firms in bankruptcy or facing other financial stress. Distressed securities strategies are usually long biased, with high illiquidity and moderate or low leverage. Returns tend to be high compared to those of other event-driven strategies.

LOS 33.d

Fixed-income arbitrage. This strategy attempts to profit from the mispricing of bonds. Sub-strategies include yield curve trades and carry trades. Fixed-income arbitrage usually uses high leverage.

Convertible arbitrage. These strategies attempt to extract “underpriced” implied volatility from long convertible bonds. Convertible arbitrage works best when there is high convertible issuance, adequate market liquidity, and moderate volatility. Liquidity issues may arise from convertibles being somewhat difficult to convert to cash. Convertible arbitrage managers typically run about 300% long and 200% short.

LOS 33.e

Opportunistic strategies tend to be highly liquid and use high leverage.

Global macro. These strategies use discretionary approaches and a range of financial instruments to exploit trends in capital markets across the world. Turbulent global markets can represent both a risk and an opportunity for global macro funds; hence the returns of these strategies can be volatile and lumpy.

Managed futures. In these strategies, a portfolio of futures contracts is actively managed using systematic approaches to provide portfolio and market diversification. Managed

futures strategies often exhibit right-tail skew during market turmoil.

LOS 33.f

Specialist hedge fund strategies operate in market niches in order to generate uncorrelated returns. Success with these strategies usually requires specialized knowledge.

Volatility traders. These strategies seek to profit from changes in the term structure of volatility. OTC options can be used to create bull spreads, bear spreads, straddles, and calendar spreads. Alternatively, other instruments including VIX futures, volatility swaps, and variance swaps can be used.

Life settlements. In these strategies, pools of life insurance contracts are purchased, and the hedge fund becomes the beneficiary. The hedge fund manager looks for policies with low surrender value, low ongoing premium payments, and high probability that the insured person will die soon.

LOS 33.g

Multi-manager hedge fund approaches use strategy diversification in an attempt to produce low-volatility, steady returns.

Funds-of-funds. This strategy comprises a hedge fund that invests in other hedge funds. Funds-of-funds can offer a very broad strategy mix but can suffer from a lack of transparency, slower tactical execution, and can also expose the FoF investor to netting risk.

Multi-strategy funds. In this hedge fund approach, a single hedge fund pursues a combination of strategies all under one roof. Compared to funds-of-funds, multi-strategy funds offer a better fee structure and faster tactical asset allocation, though operational risks are less diversified.

LOS 33.h

Conditional linear factor models can be useful for analyzing hedge fund strategies in terms of their risk factor exposures. The curriculum makes use of a specific four-factor model (incorporating equity risk, currency risk, volatility risk, and credit risk factors) to quantify a strategy's exposures.

LOS 33.i

Hedge funds generally bring diversification to traditional stock/bond portfolios, and enhance risk-adjusted returns. The addition of a 20% hedge fund allocation to a traditional 60% stock/40% bond portfolio generally decreases the portfolio's total standard deviation, increases the Sharpe and Sortino ratios, and decreases maximum drawdown.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 33.1

1. **C** Convertible bond arbitrage strategies are generally classified as relative value strategies. (LOS 33.a)

2. **A** Managed futures strategies are generally classified as opportunistic strategies. (LOS 33.a)

Module Quiz 33.2

1. **A** EMN strategies usually apply somewhat high levels of leverage in order to produce meaningful levels of return. Neither dedicated short strategies nor short-biased strategies typically make significant use of leverage. (LOS 33.b)
2. **C** Equity long/short strategies typically have gross exposures of 70% to 90% long and 20% to 50% short. Dedicated short strategies are usually 60% to 120% short at all times. Short-biased strategies are usually around 30% to 60% net short. (LOS 33.b)
3. **A** Compared to various other hedge fund approaches, EMN strategies generally have relatively modest returns. EMN funds' primary source of return is alpha. They do not take on beta risk. Their lack of market exposure make EMN strategies attractive in periods of market weakness. (LOS 33.b)
4. **A** While short positions are possible in distressed securities investing, this strategy is usually long biased. Illiquidity tends to be high, and the strategy generally uses moderate to low leverage. (LOS 33.c)
5. **A** When a firm's assets are sold off in liquidation, securities holders are paid sequentially depending on the priority of their claims: first senior secured debt, then junior secured debt, unsecured debt, convertible debt, preferred stock, and lastly common stock. (LOS 33.c)
6. **A** Convertible arbitrage managers typically attempt to extract underpriced implied volatility from holdings of long convertible bonds. To delta and gamma hedge these exposures, the managers will take *short equity* positions. (LOS 33.d)

Module Quiz 33.3

1. **C** Managed futures strategies are usually implemented via systematic approaches, while global macro strategies more often use discretionary approaches. Both strategies typically use high leverage and tend to be highly liquid. (LOS 33.e)
2. **A** Returns of managed futures and global macro strategies both typically exhibit right-tail (positive) skewness during times of market stress. Global macro strategies, however, generally deliver more heterogeneous outcomes. (LOS 33.e)
3. **C** Equity volatility is roughly 80% negatively correlated with equity market returns: volatility levels rise when equity markets fall. This characteristic makes long volatility strategies useful diversifiers for long equity investments. (LOS 33.f)
4. **C** In implementing life settlement strategies, a hedge fund manager looks for policies with the following traits: low surrender value being offered to the insured individual, low ongoing premium payments required of the investor, and high probability that the insured person will die sooner than predicted by actuarial methods. (LOS 33.f)
5. **C** Funds-of-funds generally offer a more diverse strategy mix than do multi-strategy funds. Multi-strategy funds offer *quicker* tactical asset allocation and generally a

better fee structure (for example, netting risk between strategies is often absorbed by the multi-strategy general partner). (LOS 33.g)

6. **B** Compared to multi-strategy funds, funds-of-funds offer an investor less transparency and higher netting risk. Multi-strategy funds exhibit higher variance due to less diverse strategies and use comparatively higher leverage. (LOS 33.g)

Module Quiz 33.4

1. **C** This reading uses a model that incorporates four factors: equity risk, credit risk, currency risk, and volatility risk. (The interest rate risk “BOND” and commodity risk “COMDTY” factors used by Hasanhodzic and Lo were dropped due to multicollinearity issues.) (LOS 33.h)
2. **C** Adding a 20% hedge fund allocation to a traditional 60%/40% portfolio usually *decreases* total portfolio standard deviation while it increases Sharpe and Sortino ratios in the combined portfolios. An allocation to hedge funds often *decreases* maximum drawdown. (LOS 33.i)
3. **A** Adding an allocation to equity market-neutral hedge fund strategies to a traditional portfolio has been shown to be effective in generating superior risk-adjusted performance, as evidenced by high Sharpe and Sortino ratios. (The same is true of systematic futures, global macro, and event-driven strategies.) On the other hand, fund-of-funds and multi-strategy funds have been found not to enhance risk-adjusted performance significantly. (LOS 33.i)

Topic Quiz: Alternative Investments

You have now finished the Alternative Investments topic section. Please log into your Schweser online dashboard and take the Topic Quiz on this section. The Topic Quiz provides immediate feedback on how effective your study has been for this material. Questions are more exam-like than typical Module Quiz or QBank questions; a score of less than 70% indicates that your study likely needs improvement. These tests are best taken timed; allow three minutes per question.

FORMULAS

Fixed Income

price of a T-period zero-coupon bond

$$P_T = \frac{1}{(1 + S_T)^T}$$

forward price (at $t = j$) of a zero-coupon bond maturing at $(j+k)$

$$F_{(j,k)} = \frac{1}{[1 + f(j,k)]^k}$$

forward pricing model

$$P_{(j+k)} = P_j F_{(j,k)}$$

Therefore:

$$F_{(j,k)} = \frac{P_{(j+k)}}{P_j}$$

forward rate model

$$[1 + S_{(j+k)}]^{(j+k)} = (1 + S_j)^j [1 + f(j,k)]^k$$

or

$$[1 + f(j,k)]^k = [1 + S_{(j+k)}]^{(j+k)} / (1 + S_j)^j$$

swap spread

$$\text{swap spread}_t = \text{swap rate}_t - \text{Treasury yield}_t$$

TED spread

$$\text{TED spread} = (3\text{-month MRR}) - (3\text{-month T-bill rate})$$

MRR-OIS spread

$$\text{MRR-OIS spread} = \text{MRR} - \text{"overnight indexed swap" rate}$$

portfolio value change due to level, steepness, and curvature movements

$$\frac{\Delta P}{P} \approx -D_L \Delta x_L - D_S \Delta x_S - D_C \Delta x_C$$

callable bond

$$V_{\text{call}} = V_{\text{straight}} - V_{\text{callable}}$$

puttable bond

$$V_{\text{puttable}} = V_{\text{straight}} + V_{\text{put}}$$

$$V_{\text{put}} = V_{\text{puttable}} - V_{\text{straight}}$$

$$\text{effective duration} = ED = \frac{BV_{-\Delta y} - BV_{+\Delta y}}{2 \times BV_0 \times \Delta y}$$

$$\text{effective convexity} = EC = \frac{BV_{-\Delta y} + BV_{+\Delta y} - (2 \times BV_0)}{BV_0 \times \Delta y^2}$$

convertible bond

minimum value of convertible bond
= greater of conversion value or straight value

$$\text{market conversion price} = \frac{\text{market price of convertible bond}}{\text{conversion ratio}}$$

market conversion premium per share
= market conversion price – stock's market price

$$\text{market conversion premium ratio} = \frac{\text{market conversion premium per share}}{\text{market price of common stock}}$$

$$\text{premium over straight value} = \left(\frac{\text{market price of convertible bond}}{\text{straight value}} \right) - 1$$

callable and puttable convertible bond value = straight value of bond
+ value of call option on stock
– value of call option on bond
+ value of put option on bond

credit analysis

recovery rate = percentage of money received upon default of the issuer

loss given default (%) = 100 – recovery rate

expected loss = probability of default × loss given default

present value of expected loss
= (value of risk-free bond) – (value of credit-risky bond)

upfront premium % (paid by protection buyer)
≈ (CDS spread – CDS coupon) × duration

price of CDS (per \$100 notional) ≈ \$100 – upfront premium (%)

profit for protection buyer ≈ change in spread × duration × notional principal

Derivatives

forward contract price (cost-of-carry model)

$$FP = S_0 \times (1 + R_f)^T$$

or

$$S_0 = \frac{FP}{(1 + R_f)^T}$$

no-arbitrage price of an equity forward contract with discrete dividends

$$FP(\text{on an equity security}) = (S_0 - PVD) \times (1 + R_f)^T$$

$$FP(\text{on an equity security}) = \left[S_0 \times (1 + R_f)^T \right] - FVD$$

value of the long position in a forward contract on a dividend-paying stock

$$V_t(\text{long position}) = [S_t - PVD_t] - \left[\frac{FP}{(1 + R_f)^{(T-t)}} \right]$$

price of an equity index forward contract with continuous dividends

$$FP(\text{on an equity index}) = S_0 \times e^{(R_f^c - \delta^c) \times T} = (S_0 \times e^{-\delta^c \times T}) \times e^{R_f^c \times T}$$

where:

R_f^c = continuously compounded risk-free rate

δ^c = continuously compounded dividend yield

forward price on a coupon-paying bond

$$FP(\text{on a fixed income security}) = (S_0 - PVC) \times (1 + R_f)^T$$

or

$$= S_0 \times (1 + R_f)^T - FVC$$

value prior to expiration of a forward contract on a coupon-paying bond

$$V_t(\text{long position}) = [S_t - PVC_t] - \left[\frac{FP}{(1 + R_f)^{(T-t)}} \right]$$

price of a bond futures contract

$$FP = [(full\ price)(1 + R_f)^T - FVC - AI_T]$$

quoted bond futures price based on conversion factor (CF)

$$QFP = FP/CF = [(full\ price)(1 + R_f)^T - FVC - AI_T] \left(\frac{1}{CF} \right)$$

swap fixed rate

$$SFR(\text{periodic}) = \frac{1 - \text{final discount factor}}{\text{sum of discount factors}}$$

$$\begin{aligned} \text{swap fixed rate(annual)} \\ = SFR(\text{periodic}) \times \text{number of settlements periods per year} \end{aligned}$$

value of plain vanilla interest rate swap (to payer) after inception

$$\text{value to the payer} = \sum DF \times (SFR_{\text{New}} - SFR_{\text{Old}}) \times \frac{\text{days}}{360} \times \text{notional principal}$$

probability of an up-move or down-move in a binomial stock tree

$$\begin{aligned} \pi_U = \text{probability of an up move} &= \pi_U = \frac{1 + R_f - D}{U - D} \\ \pi_D = \text{probability of a down move} &= (1 - \pi_U) \end{aligned}$$

put-call parity

$$S_0 + P_0 = C_0 + PV(X)$$

put-call parity when the stock pays dividends

$$P_0 + S_0 e^{-\delta T} = C_0 + e^{-rT} X$$

dynamic hedging

$$\text{number of short call options needed to delta hedge} = \frac{\text{number of shares hedged}}{\text{delta of call option}}$$

$$\text{number of long put options needed to delta hedge} = - \frac{\text{number of shares}}{\text{delta of the put option}}$$

change in option value

$$\Delta C \approx \text{call delta} \times \Delta S + \frac{1}{2} \text{gamma} \times \Delta S^2$$

$$\Delta P \approx \text{put delta} \times \Delta S + \frac{1}{2} \text{gamma} \times \Delta S^2$$

option value using arbitrage-free pricing portfolio

$$C_0 = hS_0 + \frac{(-hS^- + C^-)}{(1 + R_f)} = hS_0 + \frac{(-hS^+ + C^+)}{(1 + R_f)}$$

$$P_0 = hS_0 + \frac{(-hS^- + P^-)}{(1 + R_f)} = hS_0 + \frac{(-hS^+ + P^+)}{(1 + R_f)}$$

BSM model

$$C_0 = S_0 e^{-\delta T} N(d_1) - e^{-rT} X N(d_2)$$

$$P_0 = e^{-rT} X N(-d_2) - S_0 e^{-\delta T} N(-d_1)$$

Alternative Investments

Theory of Storage

commodity futures price = spot price + storage costs – convenience yield

NCREIF Property Index (NPI) calculation

$$\text{return} = \frac{\text{NOI} - \text{capital expenditures} + (\text{end market value} - \text{beg market value})}{\text{beginning market value}}$$

net asset value approach to REIT share valuation

$$\begin{aligned} & \text{estimated cash NOI} \\ & \div \text{assumed cap rate} \\ & = \text{estimated value of operating real estate} \\ & + \text{cash and accounts receivable} \\ & - \text{debt and other liabilities} \\ & = \text{net asset value} \\ & \div \text{shares outstanding} \\ & = \text{NAV/share} \end{aligned}$$

price-to-FFO approach to REIT share valuation

$$\begin{aligned} & \text{funds from operations (FFO)} \\ & \div \text{shares outstanding} \\ & = \text{FFO/share} \\ & \times \text{sector average P/FFO multiple} \\ & = \text{NAV/share} \end{aligned}$$

price-to-AFFO approach to REIT share valuation

$$\begin{aligned} & \text{funds from operations (FFO)} \\ & - \text{non-cash rents:} \\ & - \text{recurring maintenance-type capital expenditures} \\ & = \text{AFFO} \\ & \div \text{shares outstanding} \\ & = \text{AFFO/share} \\ & \times \text{property subsector average P/AFFO multiple} \\ & = \text{NAV/share} \end{aligned}$$

discounted cash flow approach to REIT share valuation

$$\begin{aligned} & \text{value of a REIT share} \\ & = \text{PV}(\text{dividends for Years 1 through } n) + \text{PV}(\text{terminal value at the end of Year } n) \end{aligned}$$

conditional factor risk model:

$$\begin{aligned} (\text{Return on HF})_t &= \alpha_i + \beta_{i,1}(\text{Factor 1})_t + \beta_{i,2}(\text{Factor 2})_t + \dots + \beta_{i,K}(\text{Factor K})_t + \\ & D_t \beta_{i,1}(\text{Factor 1})_t + D_t \beta_{i,2}(\text{Factor 2})_t + \dots + D_t \beta_{i,K}(\text{Factor K})_t \\ & + (\text{error})_{i,t} \end{aligned}$$

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